

Multi-Agent Multimodal News Verification and Explainable Detection System

PHASE I REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY: CHENNAI 600 025**

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BONAFIDE CERTIFICATE

Certified that this Report titled “**MULTI-AGENT MULTIMODAL NEWS VERIFICATION AND EXPLAINABLE DETECTION SYSTEM**” is the Bonafide work of **PRASANNAKUMAR P (230701238), PRIYANS RAJ BHANDARA (230701403), RAKESH DAVID DINESHAN JOSEPH (230701256), SUBRAMANIAN N (230701348)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal(s):

Goal No: 16

Description: Promote Peaceful and inclusive societies, provide access to justice for all and build accountable institutions for sustainable development.

Goal No: 9

Description: Build resilient infrastructure, promote sustainable industrialization and foster innovation.

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ABSTRACT

The fast diffusion of digital media through social platforms and online publishing has led to an unprecedented increase in speed and volume of misinformation. Current digital misinformation not only involves fake text but often exploits real media by editing, manipulating or even combining it with baseless information to create a pseudo reality. Monolithic and rigid classification systems are bound to be inadequate for the current task for reasons like knowledge cutoff and opaque decision-making process. In order to mitigate the above-mentioned weaknesses, this project proposes an approach which is based on Multi-Agent Multimodal News Verification and Explainable Detection System that takes social media post URLs, digital news URLs or raw media payloads as inputs and independently verifies them. This framework is comprised of autonomous agentic retrieval, NLI (Natural Language Inference), image forensics, cross-encoder evidence reranking and XAI modules. The system takes in raw text, breaks body paragraphs down into atomic statements, performs coreference resolution and activates automated web search agents to find the evidence from the real world. Candidate documents get processed through two-stage vector search and BGE Reranker reranking. A fine-tuned DeBERTa-v3 NLI model then performs the task of textual inference and outputs the assertions as Supported, Contradicted or Insufficiently Verified. The input images are detected and aligned, and then processed through a multi-stream neural network that consists of spatial feature extraction (EfficientNet-B4/Vision Transformers) in conjunction with frequency domain feature extraction (Error Level Analysis/Noise Fingerprinting). The spatial high-level features are merged with low-level noise map features for producing an authenticity image score, pixel-level image tampering localization and polygonal regions with splicing and/or synthesis tampering. In order to combine predictions without having to rely on any fixed weightage values, the Dynamic Context Engine analyzes the information density in various modalities and assigns weights to both the text and image confidence measures. Moreover, there is also a Secondary Guarantee Checker which conducts reverse cross-modal matching checks to identify “pseudo-real news.” The integrated system logs forensic maps, verification reports and decision outputs in a master database layer for fast retrieval.

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LIST OF ABBREVIATIONS

NLI	Natural Language Inference
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
FAISS	Facebook AI Similarity Search
ELA	Error Level Analysis
FFT	Fast Fourier Transform
PRNU	Photo-Response Non-Uniformity
XAI	Explainable Artificial Intelligence
DFD	Data Flow Diagram
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
BGE	BAAI General Embedding
ViT	Vision Transformer
CNN	Convolutional Neural Network
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 General

Internet has changed news gathering from a process controlled primarily by publishers to an information environment where anyone can produce, amplify, and distribute information instantly. Although this change has increased accessibility to information around the globe, it has raised difficult technical issues associated with the spread of misinformation, state-sponsored disinformation, fake news generated using artificial intelligence, misleading captions, and false narratives. Digital publishing tools, deep learning networks, and social media bots allow the perpetrators to conduct massive deception campaigns instantly.

The majority of existing approaches for identifying fake news tend to view the task of verification as a monolithic problem, which involves a binary or multiclass classification process. These approaches use machine learning models that have been trained on offline datasets (for example, LIAR, ISOT), containing labeled information in order to identify certain features of language or style to predict if an article or message is factual or not. However, the factuality of the information is not a static feature. It is rather dynamic and highly dependent on the context, location, source and external evidences. An example of a fact may be no longer true after some period of time, and an unedited photo can be misleading about some other event.

Benchmark research datasets like FEVER (Fact Extraction and VERification) and AVeriTeC (Automated Verification of Real-World Claims) show that it is necessary to search for external evidence in order to verify the claims. FEVER laid down the foundation that fact verification involves searching for evidence passages before making truth value predictions. AVeriTeC advanced this methodology by showing that it was necessary to do the same even in open-web scenarios.

On the other hand, digital misinformation in contemporary times often involves visual components. A picture attached to a news article may either be digitally manipulated, artificially produced through diffusion modeling, or genuinely true but irrelevant to the context of the story. It is thus necessary to analyze digital news veracity using:

Content-level textual verification via autonomous evidence retrieval and Natural Language

Inference (NLI), Media-level image forensics via spatial pixel inspection and frequency-domain artifact analysis, and Cross-modal contextual verification to ensure genuine visual assets actually support the specific assertions made in the text.

1.2 Existing System

The existing systems for detection of fake news can generally be classified as either manual approaches, static machine learning classifiers, evidence-based text verification pipelines, reverse image search engines, or image forensics tools.

Manual Fact Checking

Professional fact checkers verify claims, source out relevant information, corroborate findings using primary literature, analyze data, and release their conclusions through websites such as PolitiFact, Snopes, and Poynter.

Advantages: High level of human context awareness, capability to examine intricate political statements, excellent evidence selection, and sophisticated reasoning.

Limitations: Very time-consuming, hard to scale, needs domain experts to be available, and unable to perform real-time analysis on millions of posts per day on social media sites.

Machine-Learning-Based Fake News Classification

Traditional automatic systems use supervised machine learning or deep learning classifiers based on datasets (like LIAR, ISOT, and Kaggle dataset), which have been annotated offline to recognize linguistic features of false news.

Typical Approaches: Logistic Regression, Support Vector Machines (SVM), Random Forests, 1D-CNNs, Long Short-Term Memory (LSTM) networks, Capsule Networks, and fine-tuned Transformer classifiers (e.g., BERT, RoBERTa).

Limitations: The key failing here is that these models learn their patterns based on past training data. They cannot confirm an emerging breaking news story, for which no past examples existed, and thus suffer immensely from temporal decay and knowledge cutoff problems.

Text-Based Fact Verification

Fact verification approaches search for evidence in the form of external documents to check if these pieces of evidence agree or disagree with the claim made. The FEVER benchmark was proposed by the Fact Extraction and VERification approach, whereas the AVeriTeC benchmark extended this task towards open web evidence.

Limitations: It should be noted that the majority of text-based systems of verification are

unimodal, which means that such systems are unable to verify visual objects because they do not pay attention to visuals.

Image Reverse Search

Reverse Image Search engines (Google Reverse Image Search and TinEye are a few examples) search for images based on visual content within massive database repositories on the web.

Limitations: Visual similarity alone does not establish factual veracity. It cannot determine:

1. Whether an image was locally manipulated or spliced,
2. Whether the image actually belongs to the claimed event,
3. Whether the photograph was taken at the claimed date and location, or
4. Whether the accompanying textual headline is truthful.

Image Deepfake and Manipulation Detection

Deep-learning-based image forensics methods perform an analysis of visual data to detect artifacts caused by synthetic generation or pixel manipulation. Systems like TruFor fuse high-level spatial RGB features along with low-level camera noise fingerprints (Noiseprint) to provide global image integrity assessment and localization.

Limitations: Forensic image analysis determines media authenticity (is the image edited?), but it cannot evaluate narrative correctness (is the written story true?). A genuine, unedited photograph can still be used to create false news when paired with a misleading caption.

Table 1.2: Limitations in Existing Systems

Existing Approach	Primary Operational Mechanism	Major Technical Limitation
Manual Fact-Checking	Human expert investigation and source checking	Exceptionally slow and impossible to scale for real-time web streams.
Fake-News Classifiers	Supervised learning on historical text datasets	Requires pre-labeled data and becomes obsolete on breaking news events.
Text-Only Verification	Evidence passage retrieval and NLI entailment	Cannot evaluate visual media manipulation or deepfake images.

Reverse Image Search	Visual hashing and feature-matching index search	Finds visually similar images but fails to establish factual context.
Deepfake Detectors	Spatial CNN/ViT and pixel noise analysis	Determines media authenticity but fails to assess narrative truthfulness.
Multimodal Classifiers	Static feature fusion across text and images	Functions as an opaque black box, offering no evidence or citations.
Search-Based Systems	Unfiltered web query and document fetching	Retrieves low-quality or irrelevant evidence without dense reranking.

To address these limitations, a unified architecture is required that integrates real-time agentic evidence retrieval, Natural Language Inference (NLI), dual-stream image forensics, pixel-level manipulation localization, dynamic score fusion, and cross-modal contextual reasoning ("pseudo-real news" detection).

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

The ongoing development of digital disinformation necessitates a detailed study of methods for historical baseline creation, modern approaches to retrieval-augmented fact-checking techniques, visual forensics, and natural language inference architecture. The following subsections examine the most important works related to text-based claim verification, multimodal misalignment, low-level visual forensics, dense evidence retrieval, and classical deep learning baselines.

2.1.1 Textual Fact-Checking Corpora & Benchmarks

Automated textual fact-checking has evolved from simple stance classification on isolated claims to multi-stage evidence retrieval and entailment verification across complex web documents.

Wang (2017) came up with the LIAR database, which set the tone of one of the early benchmark corpora in political fact checking. The LIAR dataset is made up of roughly 12,800 manually labelled short statements taken from PolitiFact, grouped into a fine-grained six-level veracity scale (pants-on-fire, false, barely-true, half-true, mostly-true, and true). Although the LIAR dataset illustrated the weaknesses of the surface-text classifiers with feature engineering, the shortcomings of the LIAR dataset emanate from the fact that the LIAR dataset is made up of short and stand-alone political quotes without the context.

FEVER (Fact Extraction and VERification), developed by Thorne et al. (2018), is a groundbreaking dataset that consists of 185,445 claims that have been derived by mutation of sentences from Wikipedia. Such claims are assessed on the basis of three classes, namely, Supported, Refuted, and NotEnoughInfo, and for each claim, specific evidence sentences are identified in the Wikipedia dump in order to predict the validity of the claim. The fundamental concept of the FEVER dataset revolutionized fact verification tasks from simple static classification to a two-step process, including evidence retrieval and natural language inference (NLI). Our Text Verification Agent has been designed based on the evidence-first approach of FEVER.

Schlichtkrull et al. (2023) presented AVeriTeC, a rich real-world dataset for fact-checking that includes 4,568 claims collected from fact-checking agencies across the world. In contrast to synthetic datasets, AVeriTeC offers evidence-answer pairs together with multi-sentence explanations about how extracted evidence led to a certain claim classification.

AVeriTeC addresses the key problems of automatic fact-checking systems, namely, the problem of temporal leakage (when models get access to the future data during the training phase) and lack of proper evidence management. These problems have direct implications for the implementation of our system's search engine.

2.1.2 Multimodal Misinformation & Cross-Modal Misalignment

Modern digital misinformation rarely exists in a single modality. Instead, deceptive campaigns strategically fuse textual statements with visual elements to maximize emotional engagement and perceived authenticity.

The Multimodal Automated Fact-Checking (Alam et al., 2022; Gullapalli et al., 2023) stresses that online misinformation makes use of "out-of-context" (OOC) asset pairs instead of generating synthetic images. In such cases, a completely real and unaltered image is used to back up a fake textual claim (i.e., using an actual 2018 flood image to illustrate a current incident in another country). Human cognition causes individuals to give credence to posts having visuals. Therefore, the reality of images adds to the authenticity of the fake texts. This requires a clear distinction between:

- **Media Authenticity** (Is the image synthetically generated or digitally edited?),
- **Claim Factuality** (Is the textual assertion factually correct based on current real-world data?), and
- **Contextual Consistency** (Does the visual asset actually depict the specific event described in the text?).

Fakeddit was introduced by Nakamura et al. (2020) to be an enormous multimodal dataset comprising of more than a million fine-grained posts containing text and images from Reddit. Fakeddit offers a multi-level taxonomy categorizing fine-grained samples into manipulated text, false context, look-alike media, and complete fabrication. The Fakeddit research showed that multimodal feature representations always beat unimodal text and image classifiers. The current project is based on the Fakeddit dataset by incorporating multimodal feature representation with agentic web search and clear language LLM explanations instead of black box classification outputs.

2.1.3 Image Forensics & Low-Level Artifact Detection

Detecting visual manipulation requires combining high-level semantic analysis (e.g., face boundaries, lighting consistency) with low-level spatial and frequency-domain artifact analysis.

Guillaro et al. (2023) proposed TruFor, an advanced image forgery detection and localization method. This method uses high-level spatial features of an image in the RGB

domain obtained using deep neural network architecture, as well as low-level noise features obtained using high-pass filters. This method generates a global image integrity score, as well as a dense manipulation localization map and a confidence reliability map at the pixel level. The visual detection approach that we propose uses this hybrid approach, where CNNs and ELA are used in tandem.

Noiseprint was introduced by Cozzolino and Verdoliva (2019) as a deep learning-based method for extracting camera model signatures and pattern noise (PRNU). By eliminating the semantic content and emphasizing the noise created by the sensors, Noiseprint is able to identify the spliced parts of the images that are generated using different cameras. This confirms the importance of adding low-level noise mapping as another stream to our framework next to semantic vision models.

The efficient compound scaling of EfficientNet was proposed by Tan and Le (2019), where network depth, width, and input image resolution are scaled using scaling factors defined by a constant factor. In our visual hybrid stream, the EfficientNet-B4 is used as the spatial feature extractor, providing the best balance in terms of efficiency and detailed feature extraction while searching for pixel modifications and boundaries.

2.1.4 Language Models & Evidence Retrieval Architecture

Evaluating textual entailment between extracted web evidence and user claims requires language models optimized for Natural Language Inference (NLI) alongside neural reranking systems.

He et al. (2021) have developed DeBERTa-v3 (Decoding-enhanced BERT with Disentangled Attention) that improves upon regular transformer architectures by utilizing two vector representations (content vector and relative position) for words and replacing the concept of Masked Language Modeling (MLM) with the RTD (Replaced Token Detection). DeBERTa-v3 has proven its excellent performance on NLI datasets (such as MNLI and FEVER). In the pipeline described above, the task of NLI is performed using fine-tuned DeBERTa-v3 cross-encoder that determines whether the retrieved web passages entail, contradict or are neutral to the extracted text.

Xiao et al. (2023) presented BGE Reranker (BAAI General Embedding), illustrating the power of the two-stage retrieval approach for open-domain question answering and claim verification tasks. Although the fast bi-encoder dense vector search (e.g., FAISS) is capable of choosing the top- K documents from the web, the bi-encoder lacks the ability to capture the interaction between fine-grained tokens. Using a cross-encoder like BGE Reranker for filtering in the second stage helps to choose only the contextually relevant evidence.

2.1.5 Classical Deep Learning & Moderation Baselines

To contextualize the performance of modern Transformer and agentic retrieval architectures, classical deep learning moderation models provide key baselines regarding computational efficiency and parameter trade-offs.

The research carried out by Saeed et al. (2018) to test multi-label classification techniques using 1D-CNN, GRU and LSTM with pre-trained FastText embeddings as initial vectors has shown that spatial dropout helps avoid hidden-layer co-adaptation and focal loss can be used for class imbalance problems. These models have reached 90% accuracy but require offline training on structured data sets and cannot perform real-time evaluation of news without integrating the search function.

The approach put forward by Ibrahim et al. (2018) is the ensemble deep learning framework using Bi-GRU, CNN, and LSTM models. Class imbalance was resolved through the use of textual data augmentation which includes synonyms replacement, paraphrase generation, and random token deletion. Although useful in closed domain text moderation, the system is not adaptable to slang changes or images.

Capsule Networks (CapsNets) was introduced by Srivastava et al. (2018) as a novel framework aimed at replacing conventional max pooling layers with dynamic routing and nonlinear squashing methods. Thus, the model could maintain spatial hierarchies and orders of tokens in datasets composed of different languages. Nevertheless, the network is still computationally expensive and can process only static textual inputs.

The study by Maslej-Krešňáková et al. (2020) compared pre-processing techniques, various types of embeddings such as TF-IDF, GloVe, FastText, and Word2Vec, and neural network models including different versions of BERT, CNNs, and Bi-LSTMs. The results showed that TF-IDF worked well with structured pre-processing, while language models such as BERT performed better with unstructured data. Unfortunately, BERT model requires a lot of computational power and cannot be accessed via live search.

Invariant Rationalization by Chuang et al. (2021) was devised as a method to remove any spurious relationships or demographic biases in text classifiers. The use of causative invariances helped in lowering the number of false positives in the text classification model. Even though the model works perfectly well for text classification, it ignores any other forms of media.

Guillaume et al. (2022) & Duchene et al. (2023) conducted a study by comparing Transformer models (BERT, RoBERTa, DistilBERT, BERTweet) with traditional CNN-BiGRU baseline models. They observed that RoBERTa trained using focal loss resulted in

state-of-the-art ROC-AUC performance while mitigating identity bias. However, it was pointed out by the authors that frozen Transformer weights can still be susceptible to reality after training due to factual information on the web.

2.2 Comparison and Discussion

Table 2.2: Comparison of Existing Systems

Study/System	Text	Image	Evidence Retrieval	Localization	Explainability	Multimodal
LIAR	✓	✗	Limited	✗	Limited	✗
FEVER	✓	✗	✓	✗	Evidence-based	✗
AVeriTeC	✓	✗	✓	✗	✓	✗
Fakeddit	✓	✓	Limited	✗	Limited	✓
Noiseprint	✗	✓	✗	✓	Visual	✗
TruFor	✗	✓	✗	✓	Heatmap	✗
Conventional reverse search	✗	✓	✓	✗	Limited	✗
Proposed system	✓	✓	✓	✓	✓	✓

Based on the literature, a singular technique does not fully address the issue of verifying whether a particular item of news on the internet is genuine. Text-based fact-checking entails the gathering of evidence while image-based fact-checking involves forensic examination. Additionally, a true image can be used to verify a false assertion.

Thus, the suggested system incorporates content-based verification and context-based verification.

2.3 Conclusion

The comprehensive analysis of existing literature, baseline benchmark datasets, and modern algorithmic architectures reveals three critical requirements that must be met to establish a robust, real-world fake news detection framework:

Dynamic Web Retrieval and Evidence-First Entailment: As is shown through the weaknesses of static data sets (LIAR, FEVER, AVeriTeC) and fixed models (RoBERTa,

DeBERTa-v3), fact checking requires more than just static classifier parameters. Fixed models will always be subject to the knowledge cutoff limitation and time leakage in validating novel out-of-the-box claims. It is imperative that system architecture takes up the evidence-first retrieval approach, whereby dense two-stage rerankers (BGE Reranker) and automatic retrieval agents are used to obtain live web evidence with respect to the target claim.

Decoupling Authenticity and Alignment between Modalities: Evidence from multimodal benchmark datasets (Fakeddit, Alam et al.) and visual forensics systems (TruFor, Noiseprint, EfficientNet) demonstrate that it is essential to decouple the integrity of media from the factual accuracy of the story being told. Disinformation on social networks involves the construction of "pseudo-real" elements where real, untouched images are accompanied by stories which are false in context. To build a robust disinformation detection system, it is important to verify localized pixel frequency differences (by leveraging CNN-ViT-ELA ensembles) and conduct a secondary cross-modal check of alignment.

Interpretable, Localized, and Evidence-Based Explainability: Current state-of-the-art deep learning models (such as Capsule network, 1D CNN + GRU), work like black box systems, providing only raw probability metrics ("88% Fake") which are uninterpretable and hard to trust or understand. A verification system should provide explainable results, i.e., pixel-wise localization map of manipulations on visual content, highlighting of conflicting claims from the text along with an explanation supported by citations from the web domain.

Multi-Agent Multimodal Fake News Detection System is the perfect solution to the above mentioned challenges. With its ability to integrate agents for searching information on web, Natural Language Inference (NLI), spatial and frequency based image forensics, dynamic contextual modality fusion and secondary guarantee checker in one single framework, the system successfully closes the existing operational gaps recognized in the previous research.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

Fig 3.1 shows the overall architecture of the phase 1 project. The system architecture is made up of an input ingestion module that gets web links of the articles or news and sends it to the multi-modal content retrieval modules. In phase 1, the text processor and image processor modules are planned for execution, so that is what we will be proceeding to talk about. The text module uses RAG-based claim search and NLI for content retrieval, while the image module uses ensemble methods for content processing. Triage and Dynamic Fusion balances the confidence scores using LightGBM to produce a composite authenticity score. Pseudo-Real Secondary Verification does an agentic-reverse search and earliest appearance check to catch fabrication. The Explainer agent does the job of converting the low-level model detections into human-readable verdicts with highlights, reasoning and citations. The findings are indexed in a PostgreSQL database and Qdrant is used to enable caching of verdicts for fast access of similar news.

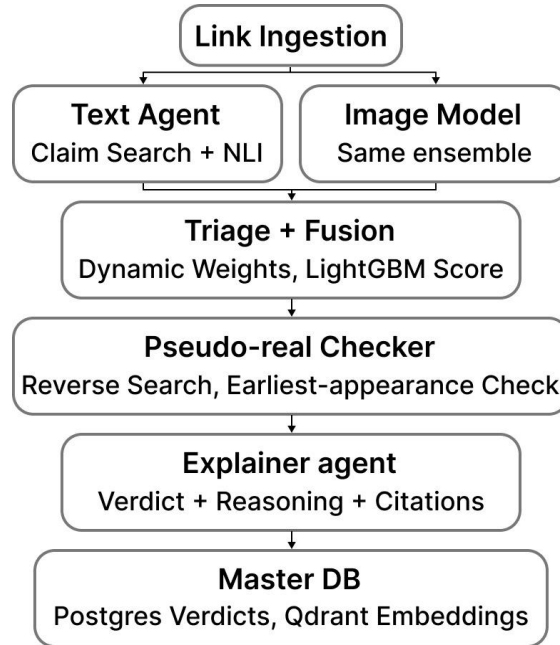


Figure 3.1: Overall architecture of the proposed multi-agent multimodal news verification system.

The system receives a link and spreads extracted information to the Text module and Image module. Their outputs are passed to a triage and fusion layer, followed by pseudo-real verification, explanation LLM and storage.

Fig 3.2 shows the text verification architecture. The presented pipeline is a well-organized

digital fact-checker functioning via five consecutive steps. First, the messy text is cleaned by replacing ambiguous pronouns with specific person's name, splitting long texts into small factual claims and tagging them with their precise locations. Then, the system searches for the evidence via complex and multi-layered algorithm that begins from its own instant database of debunked rumours, continues with queries to known fact-checking organizations and finishes with a targeted live web search. The sophisticated filter then extracts the most relevant and trustworthy sentences from search results. An inference engine analyses the found facts in comparison with the initial statement to check whether the facts confirm, disprove or leave neutral about it and even launches one more search cycle in case of necessary evidence weakness.

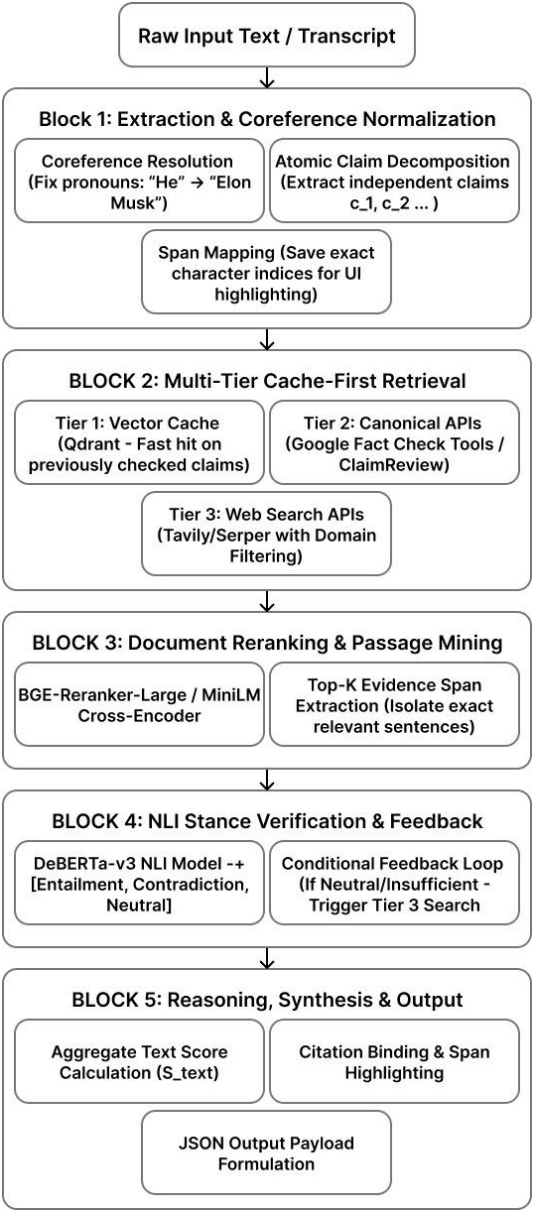


Figure 3.2: Proposed Text Verification Agent architecture.

At last, the output of the whole process is presented in the form of well-structured report with the score of claim authenticity, visual color-coded highlighting of the text, references to the sources and explanations.

Figure 3.3 shows the image module’s architecture. The described forensic process flow is a fully automated digital photo detective tool which recognizes modified pictures and provides human-understandable output about it. Firstly, the algorithm processes the loaded picture by detecting all human faces on it and cropping them in a proper manner while equalizing the size of the image for the sake of providing the consistent input for every further stage. Then it performs two independent forensic analyses of the input image. One of them is based on examining the pixels of the visual image itself while searching for any signs of the face-swapping and cut-and-paste modifications. The second process flow uses a math analysis of the hidden frequency patterns of the image in order to detect unnatural grids and artifacts of the digital image resampling. Further, the results of both analyses are combined into a unified high-precision heatmap and then the overall score of the image authenticity is calibrated.

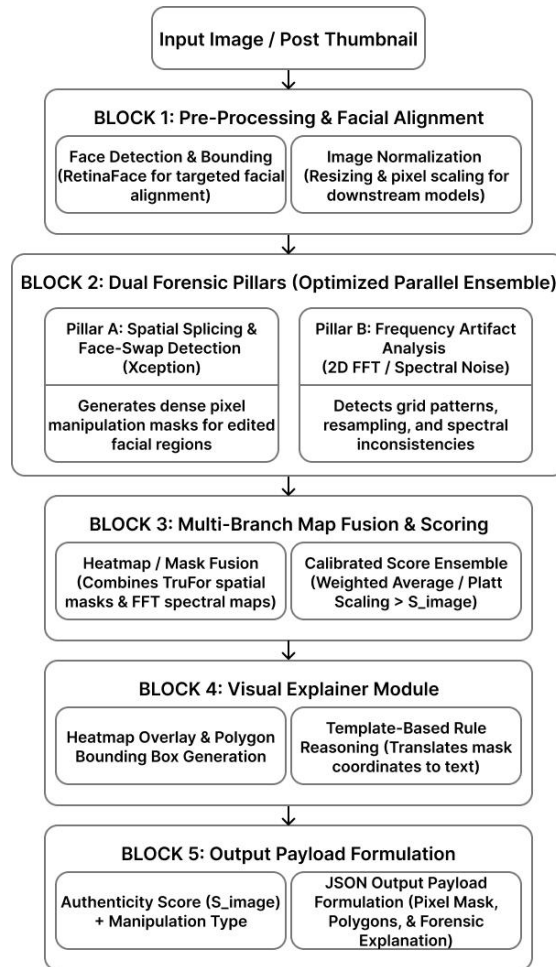


Figure 3.3: Proposed Image Forensics and Explainability architecture.

After that, the visual explainer is used for interpreting the coordinates of the anomalies and transforming them into polygons while adding human-understandable annotations like indicating which exact face was modified.

3.2 System Requirements

This section depicts the software requirements, hardware requirements, datasets and deployment & scaling requirements needed for the proposed system.

3.2.1 Software Requirements

Table 3.2.1: Software Requirements

Component	Requirement
Operating System	Windows 11
Programming Language	Python-3.12
Deep Learning Framework	PyTorch $\geq$ 2.5.1
NLP Framework	Hugging Face Transformers $\geq$ 5.1.5
Image Processing	OpenCV-4.8.0, Pillow-12.3.0
Face Detection	RetinaFace $\geq$ 0.0.18
Image Forensics	Timm-1.0.29
CNN Backbone	EfficientNet-B4
NLI	DeBERTa-v3
Reranking	BGE Reranker
Vector Database	Qdrant $\geq$ 1.17
Relational Database	PostgreSQL=18.6
Web Retrieval	Google Fact Check Tools API

Search	Tavily $\geq$ 1.0
API Layer	FastAPI $\geq$ 0.141.1
Frontend	React-19.3.0
Data Format	JSON

Qdrant fits into the retrieval module since it supports vector similarity search and metadata filter options which enable us to filter retrieved evidence according to specific criteria like source, date, category or domain.

The Fact Check Tools API provided by Google is an application programming interface for accessing fact-checked claims and includes the ability for both text-based claim searching as well as image-based claim searching.

The Web search module can further restrict searches to certain domains.

3.2.2 Hardware Requirements

Proper provisioning of compute infrastructure to facilitate smooth development, testing and deployment of deepfake detection and automated fact-checking pipeline is a key aspect. Contemporary multi-modal pipelines that involve vision foundation models, vector embeddings and dense local indexing have considerable requirements concerning not only processing bandwidth but also system memory. Although standard development setups can accommodate initial prototype development and operation using APIs, conducting local inference, frame pre-processing and vector similarity search would require specific acceleration. The table below provides minimal and recommended hardware requirements that need to be met in order to run the system. Minimal configuration would allow for enough overhead to conduct base level debugging and unit testing as well as batch testing. On the contrary, recommended configuration would provide higher VRAM and system memory capabilities in order to avoid out-of-memory constraints when conducting extended batch testing and inference of models. Additionally, use of fast solid-state storage allows for efficient I/O performance when loading weights of models and video data sets.

Table 3.2.2: Hardware Requirements

Component	Minimum Configuration	Recommended Configuration
Processor (CPU)	Modern Multi-core CPU	8+ Core High-Performance CPU
System Memory (RAM)	16 GB	32 GB
Graphics (GPU)	NVIDIA GPU (~8 GB VRAM)	NVIDIA GPU (8–16 GB VRAM)
Storage (SSD)	512 GB SSD	1 TB NVMe SSD

3.2.3 Dataset Requirements:

Text Verification

In order to rigorously benchmark and validate the automated text verification pipeline, multiple well-known fact-checking datasets provide distinct structural benefits in different verification challenges. Among the main datasets, FEVER (Fact Extraction and VERification) stands out as a benchmark dataset with 185,445 structured claims associated with evidence based on Wikipedia articles, suitable for assessing baseline retrieval and deductive reasoning abilities. In the case of open-domain and practical applications, AVeriTeC contains 4,568 realistic claims along with retrieved evidence, questions-and-answers and full-fledged natural language explanations, which can be seen as an intermediate dataset between the clean corpora of Wikipedia and chaotic web ecosystems. Moreover, the LIAR dataset includes over 12,800 short politically-based statements, extracted from PolitiFact, useful for verifying a six-point classification of truthfulness, whereas Fakeddit dataset features a large multimodal dataset with over a million text and image-based false statements taken from Reddit. Usage of a combination of these datasets, especially the usage of FEVER for testing dense semantic retrieval and AVeriTeC for open-web multi-source synthesis, can be beneficial for validation of the verification system on both synthetic and real claims.

Image Verification

In order to fully evaluate and validate the visual integrity aspect of the verification process, there are many classical as well as modern datasets that can be utilized for various manipulation types. In case of facial manipulation and fake video forensics, the FaceForensics++ dataset is a key benchmark, which includes various facial reenactment and facial swap algorithms at different compression levels. In terms of modern generative synthesis detection, the unique contribution of GenImage is a one-million-scale benchmark of more than one million pairs of real and AI-generated images for different architectures (e.g. Stable Diffusion, Midjourney, BigGAN) in order to explicitly measure the generalization power and resilience to degradation. Furthermore, CIFAKE provides an easy-to-use, standardized, synthetic-vs-real images dataset based on the CIFAR-10 set for fast prototyping and validation purposes. Finally, regarding the detection of classical digital forgeries, the classic forensic benchmarks include CASIA Image Tampering Dataset, which focuses on realistic splicing and copy-move editing, Columbia Image Splicing Dataset (uncompressed images) and COVERAGE Dataset, which includes images with copy-move manipulation having similar backgrounds.

3.3 Activity Diagram

Figure 3.3 explains the stages to activities happening through the system. The multimodal veracity verification process works by dividing the submitted content into two parallel streams one for text analysis based on fact retrieval through NLI, and another for the image analysis using spatial and frequency domain forensics and then combining the outputs from both processes through a dynamic weighting and contextual reasoning approach in order to create the final veracity decision. The submitted content is first broken down into two parallel streams: text stream goes through the processes of claim extraction, evidence retrieval, reranking and NLI verification; media content stream uses spatial, frequency and map fusion forensics as described in diagram below.

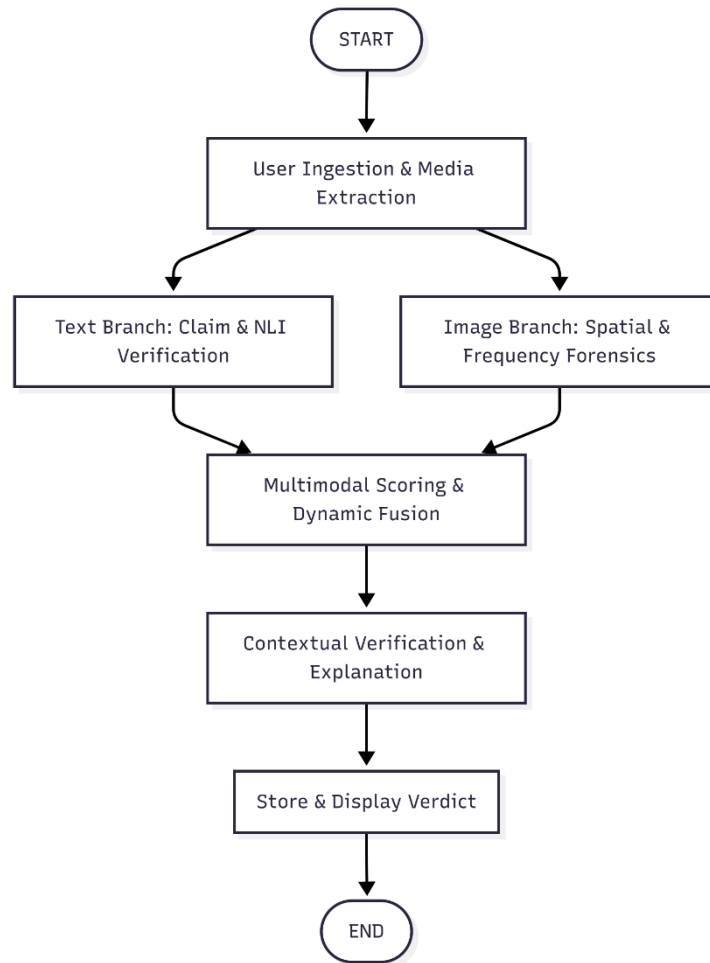


Figure 3.3: Activity Diagram

3.4 Data Flow Diagram

Figure 3.4 is the Data Flow Diagram and it displays the entire process of multimodal verification, which starts with the ingestion of the link by the user and splits into two separate pipelines for text and images. The text pipeline detects the claims, gathers the external evidence and calculates the text veracity score through Natural Language Inference (NLI). In the meantime, the image pipeline analyses the visual crops through forensic analysis and generation of the frequency/space heatmaps. Unimodal results of each pipeline will be merged in score fusion and pseudo-real context validation, in order to resolve any discrepancies between the different modalities.

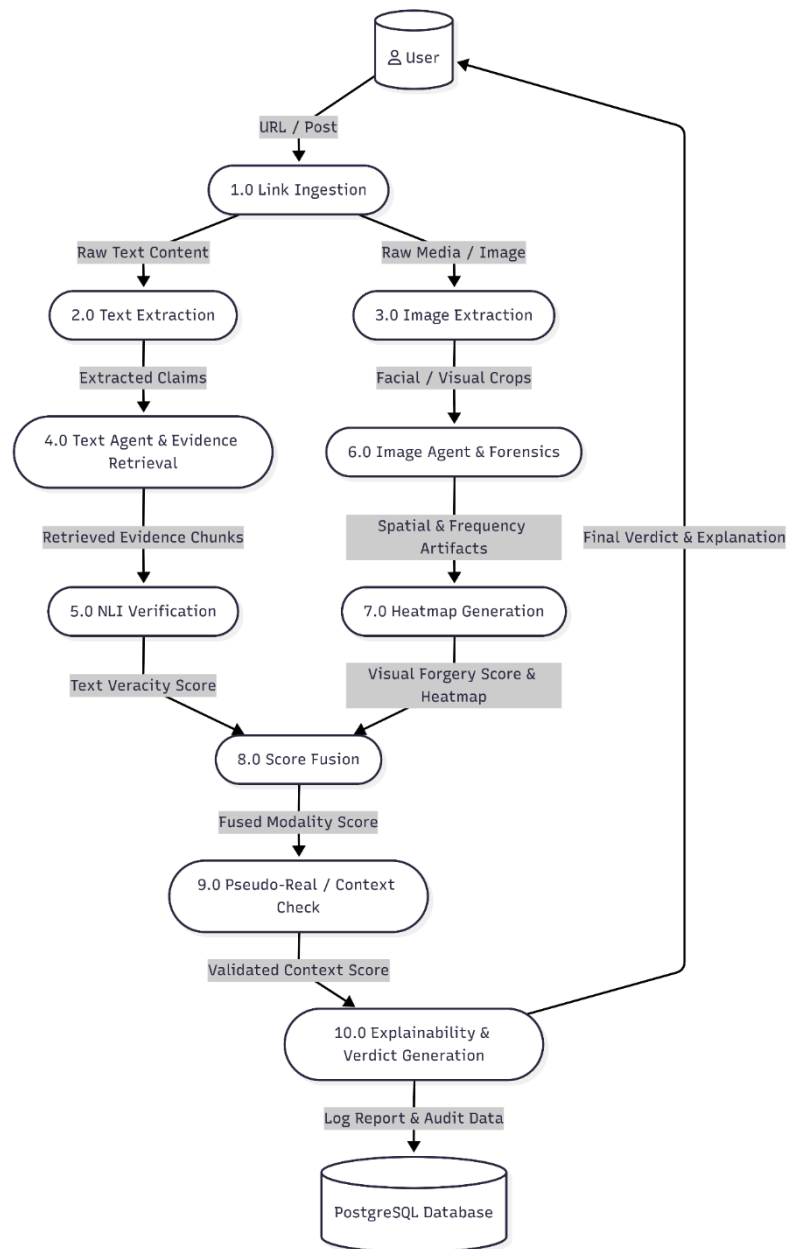


Figure 3.4: Data Flow Diagram

CHAPTER 4

PROPOSED SYSTEM

4.1 Objectives

The main aim of this suggested system is to develop an intelligent news verification system that can evaluate the credibility of the content found online and on social media by determining whether the piece of information is true, false, or unverified. This is done through a process whereby the system accepts any news article, social media post, or link as an input and extracts both text and visual data to analyze the input. Text is broken down into separately verifiable atomic claims where accurate text spans are preserved for accurate highlighting and citation binding. The system then fetches evidence either supporting or contradicting the claims from credible online sources. At the same time, the visual data is subjected to forensic examination in the spatial and frequency domain to identify the presence of image tampering as well as to create heatmap visualizations of the same. For tackling difficult instances such as pseudo-real news wherein real images are coupled with misleading stories, there is an additional step wherein the alignment between the modalities is checked. Instead of providing a binary output, the model intelligently weights the scores of each modality and combines them to provide explanations for the same.

The scope of the implementation of the current project includes the analytical and output components of the verification pipeline. These include the Text Verification Module responsible for evidence retrieval, reranking, and NLI-based textual verification, as well as the Image Forensics Module responsible for localizing image manipulations. It also includes explanation generation and citation generation ending up with structured result generation.

4.2 Methodologies

The suggested approach follows the modular agentic verification approach. Rather than considering verification as a one-classification task, the system breaks down verification into several tasks.

4.2.1 Agentic Verification

The agent oversees the whole process of searching, from deciding on which information should be searched to finding out whether more searches are required or whether the information already collected is enough to terminate the process of verification. Such a dynamic approach that involves multiple steps is critical when dealing with the processing of text, as the reality of facts keeps changing.

4.2.2 Retrieval-Augmented Verification

External evidence is gathered before giving out the verdict. This gathering is used in the process of verification.

The retrieval order is as follows:

Tier 1: Vector Cache

Verified claims and evidence are stored in the form of embeddings in Qdrant. These claims and evidences form the base of our verification.

If there is a claim that is semantically similar, it can be easily found. That is how the system easily verifies fake news.

Qdrant not only provides similarity search but also filtering based on payloads. This allows for user convenience.

Tier 2: Canonical Fact-Checking Sources

Sources such as Google Fact Check results, Claim Review and other well-known fact-checking agencies are queried by Google's API through which fact-checked claims can be found with textual queries and filtering based on publishers' site.

Tier 3: Web Retrieval

If enough evidence is not available, web retrieval is performed with the help of domain filters and others. This ensures that correct information is always available in terms of verification.

4.3 Modules

4.3.1 Link Ingestion Module

The Link Ingestion Module is responsible for receiving a user-provided URL or post and extracting available information.

Functions

- URL validation
- webpage retrieval
- article text extraction
- metadata extraction
- image extraction
- thumbnail extraction
- media identification
- duplicate detection

Output

{

```
"url": "...",  
"title": "...",  
"text": "...",  
"images": [],  
"metadata": {}  
}
```

4.3.2 Text Verification Module

This module is one of the two main modules that have been implemented. Its structure is according to the first given diagram.

A. Coreference Resolution

News texts sometimes include pronouns or references like:

“He announced the policy.”

The system needs to know who “He” is before the process of finding any evidence.

Coreference resolution converts:

He announced the policy.

to:

“Elon Musk announced the policy.”

when possible. The process is helpful as search queries with entities are usually more informative than pronouns.

B. Decomposing Atomic Claims

A sentence can have multiple verifiable claims.

Example:

"The government has rolled out the program in 2025, creating 10,000 jobs."

This can be broken down as follows:

Claim c1: Government had rolled out the program in 2025.

Claim c2: The program created 10,000 jobs.

Each claim can be independently verified.

It would not be right to verify a whole sentence based on a single claim.

C. Span Mapping

By preserving the original character positions of extracted claims—such as mapping a claim back to its exact start and end indices—the system enables the frontend interface to pinpoint precise text segments and dynamically highlight whether specific portions of the content are supported, contradicted, or uncertain.

4.3.3 Multi-Tier Retrieval

Layer 1: Qdrant Vector Cache

Historical claims are embedded and stored in a Qdrant vector database. Upon the entry of a new claim in the system, it is embedded and evaluated through similarity search against historical claims stored in the database, saving lots of redundant API calls because previous verified claims are retrieved right away.

Layer 2: Canonical Fact Checking APIs

The system queries canonical databases for evidence of prior fact-checking of claims. With the help of integration tools like Google's Fact Check Tools API, it gets the structured information about the verified claim including its publisher, URL of the review, verification date, and ratings.

Layer 3: Web Search

If vector cache and canonical APIs do not give enough evidence, the architecture relies on a web search. The third layer employs domain restriction to find trusted, high-authority resources such as government websites, reputable news websites, universities, international organizations, and fact-checking websites.

4.3.4 Evidence Reranking

Because initial retrieval often yields a broad set of candidate documents, the system applies a reranking stage using a BGE Cross-Encoder. The input claim and retrieved documents are processed through the encoder to generate precise relevance scores, filtering the candidates down to the top-K evidence passages. This process ensures the system isolates passages that directly address the core claim rather than merely selecting text with overlapping keywords.

4.3.5 NLI Verification and conditional Feedback loop

Entailment

The evidence explicitly supports the premise of the claim $P(E|C) \rightarrow \text{Supported}$. Under this condition, the retrieved text validates the factual accuracy of the claim, leading to a verdict of "**Supported.**"

Contradiction

The evidence directly conflicts with or denies the information stated in the claim $P(C|E) \rightarrow \text{Refuted}$. In this scenario, the retrieved sources disprove the assertion, resulting in a verdict of "**Refuted.**"

Neutral

The retrieved evidence neither validates nor refutes the assertion, indicating a lack of sufficient context or missing data. This outcome flags the claim as having "**Insufficient**

Evidence" and determines whether additional retrieval loops are required.

DeBERTa-v3 serves as the primary NLI model due to its advanced natural language comprehension capabilities. Rather than blindly executing maximum web searches every time, the pipeline dynamically adapts based on confidence outputs: high-confidence predictions immediately resolve into a final verdict, whereas low-confidence predictions automatically route back to Tier 3 web retrieval to gather fresh evidence for a subsequent evaluation pass.

4.3.6 Text Score Calculation

For claim c_i , define:

$$S_i = w_e E_i + w_c C_i + w_r R_i + w_s S_i^{\text{source}}$$

where:

- E_i = NLI evidence score
- C_i = evidence consistency
- R_i = retrieval relevance
- S_i^{source} = source reliability
- w_e, w_c, w_r, w_s = weights

The overall text score can be calculated as:

$$S_{\text{text}} = \frac{\sum_{i=1}^n \alpha_i S_i}{\sum_{i=1}^n \alpha_i}$$

where α_i represents the importance of claim i .

This allows important claims to contribute more strongly to the final text score.

4.3.7 Citation Binding

Every verified claim should be connected to its evidence.

The output can contain:

```
{  
  "claim": "The event occurred in 2025.",  
  "verdict": "SUPPORTED",  
  "score": 0.91,  
  "evidence": "...",  
  "source": "...",  
  "span": {  
    "start": 152,
```

```

    "end": 185
  }
}

```

This makes the system explainable and enables citation-aware highlighting.

4.3.8 Image Forensics Module

The next implemented module is the Image Forensics Module.

Its architecture is based on the second provided diagram.

A. Image Pre-processing

The image to be processed must first be normalized.

The pre-processing tasks include the following operations:

- Image decoding
- Image resizing
- Normalization
- Color space transformation

Preservation of image artifacts in pre-processing.

In case of facial manipulations, RetinaFace may be used for face detection and alignment.

4.3.9 Spatial Forensic Pillar

The spatial pillar will analyze the RGB image for signs of manipulation. The architecture used in this design involves TruFor and EfficientNet-B4.

The spatial pillar is able to produce a manipulation probability map,

$M_s(y)$

where every pixel represents the probability of that particular region being manipulated.

TruFor is very relevant in this case because of its ability to produce image-level forensics and forensic map

4.3.10 Frequency Forensic Pillar

Manipulation could result in inconsistencies which are not easily discernable in RGB space.

The image is then transformed into the frequency domain using a two-dimensional Fourier transform.

$$F(v) = \sum_x \sum_y f(y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

After that, the magnitude spectrum is checked for any anomalies. The frequency domain analysis will help in identifying:

- Resampling artifacts

- Patterns
- Unusual spectra distribution
- Compression artifacts
- Frequency anomalies

4.3.11 Multi-Branch Map Fusion

The spatial and frequency maps are combined:

$$M_{fusion} = \lambda M_s + (1 - \lambda) M_f$$

where:

- M_s = spatial forensic map
- M_f = frequency-domain map
- λ = fusion coefficient

The resulting map represents the combined manipulation likelihood.

4.3.12 Image Authenticity Score

An image authenticity score converts pixel- or patch-level manipulation probabilities into a single metric ($\text{Sim}_{\text{image}} = 1 - (1/N) \sum(x,y) M_{\text{fusion}}(x,y)$), that shifts the focus from detection to authenticity. Because simple mean pooling dilutes small, localized edits across untouched areas, weighted functions or calibration techniques—like Platt scaling, isotonic regression, or temperature scaling—are used to prioritize suspicious regions and improve score interpretability. Finally, qualitative labels ranging from "Highly suspicious" to "Highly consistent" serve as practical evaluation tools, with specific cutoffs (0.30, 0.50, 0.70, and 0.90) tuned using validation data.

4.3.13 Visual Explainability

For explainable image forensics, the task is to locate regions where the model suggests possible manipulation. The forensic model produces a heatmap of the manipulation probability at the pixel or patch level; however, such a raw heatmap is not easily understandable to the end user. It goes through a series of processing steps: Original Image -> Forensic Heatmap -> Thresholding -> Connected Components -> Polygon/Bounding Box -> Explanation. Thresholding transforms the heatmap into a binary mask by selecting only those regions with confidence scores greater than a certain threshold value, while connected component analysis combines close-by flagged pixels into a region and ignores any noise. These regions are then converted into polygons that provide more precise shapes or bounding boxes. The explanation generated needs to be probabilistic and clearly states the limitation of the model. Instead of stating “this region is manipulated,” the system could

state “this highlighted facial region is more inconsistent forensically compared to surrounding regions and has higher chances of being manipulated.” It is important to note here that although the model finds the statistical inconsistencies related to manipulation, it doesn’t know how the manipulation occurred, what the intention behind the manipulation was, or who the manipulator is.

4.3.14 Dynamic Score Fusion

The complete architecture allows the contribution of each modality to change according to the input.

For example:

$$S_{final} = w_t S_{text} + w_i S_{image} + w_v S_{video} + w_a S_{audio}$$

subject to:

$$w_t + w_i + w_v + w_a = 1$$

For the current implementation:

$$S_{final} = w_t S_{text} + w_i S_{image}$$

The weights can depend on:

- presence of each modality,
- confidence of each module,
- quality of the input,
- claim importance,
- forensic confidence,
- evidence availability.

4.3.15 Pseudo-Real News Checker

This part tackles the fundamental flaw in multimodal disinformation detection: media authenticity doesn't mean truthfulness of the narrative. In the first place, a statement may say that "a stampede happened in Chennai yesterday" and use an authentic photograph of the event that isn't edited in any way. When one verifies the textual claim and finds that the stampede indeed took place, the image will likely turn out to have been published somewhere else in the past, years ago, before the time of the claim.

The basic concept is that Media Authenticity $\neq$ Narrative Truth. The image may not be edited at all, but still, be misleading due to its misleading caption, location, date, or event association. At the same time, the photo may be edited in some cases, but the claim made using it may be true. The two characteristics are independent of each other; one can ask

separately whether the media has been digitally altered and whether it is relevant in the context in which it is used. As a result, the pseudo-real checker takes into account the factors such as the image's provenance, the first known publication date, historic captions, the claimed location and date, the depicted event, and the relation between the provenance and the new claim. This creates a two-axis verification tool: Is the media authentic? Is it used to back up the true narrative?

4.3.16 Explainability Agent

Whereas the visual explainability module provides an explanation for the forensic conclusions on an individual image basis, the Explainability Agent works at the system level by combining the outputs of all the important modules, including text verification, image forensics, and pseudo-real checker, into one coherent explanation. The key design consideration for the Explainability Agent is grounding in evidence, i.e., each statement produced by the agent should correspond to an artifact of the pipeline in question, such as evidence documents, NLI entailment/contradiction, forensic manipulation score, image metadata, and localization results. Such an approach prevents the generation of the so-called "hallucinations," when the language model produces fluent yet irrelevant explanations of the underlying models' conclusions; a convincing yet baseless explanation may be more dangerous than the absence of any explanation as it creates the illusion of truthfulness. Typical output would look like this: "Verdict: Likely Misleading. The textual claim is partially backed by the information found from the sources. Nonetheless, the attached picture provides no indication of any kind of digital tampering. Reverse search results prove that the image has been previously published under different circumstances. It means that the picture can be considered authentic but it has been used in order to prove another story." It clearly shows how important it is to keep the distinction made by the pseudo-real checker, because the analysis explains the authenticity of the image and its contextual relevance separately.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 Implementation and Results

This document details the implementation of the Text Verification Module in DeepShield, following the architectural specifications for pre-processing, extraction, retrieval, and NLI verification.

5.1.1 Text Pre-processing

Conversion of raw unstructured text into structured text for atomic claim extraction and high precision retrieval forms the first step of the pipeline.

Operations

The system performs the following preprocessing pipeline operations:

HTML & Noise Removal: Cleaning of input text from markup and other noise.

Unicode Normalization: Normalizing the characters so that matching is consistent for retrieval.

Coreference Resolution: Pronouns such as “He”, “It”, and “The Project” are recognized using LLMCorefResolver and replaced by their actual entities. This is done so that queries for retrieval are complete and consistent.

Claim Decomposition: Decomposing compound sentences into “Atomic Claims” which are individual factual claims.

Span Tracking: The system keeps track of `character_start` and `character_end` of each of the extracted claims.

Example Transformation

Original Input: *"Elon Musk announced the project. He said it would create 10,000 jobs."*

Normalized (via Coreference Resolution): *"Elon Musk announced the project. Elon Musk said the project would create 10,000 jobs."*

Atomic Claims:

- **C1:** "Elon Musk announced the project." (Span: 0–32)
- **C2:** "Elon Musk said the project would create 10,000 jobs." (Span: 34–82)

5.1.2 Atomic Claim Extraction

Figure 5.1.2 shows the system adopting a well-defined process of LLM Claim Extraction to transform the paragraphs into independent verifiable claims, which are then represented using the Claim Pydantic model.

The LLMClaimExtractor operates under a strict schema and, therefore, for every claim

extracted, the system collects the following details: Claim Text: The exact substring in the original text. Character Offsets: Start and End positions of the claim. Confidence: Confidence level of the claim’s atomicity. Entities: Entities identified (PERSON, ORG, GPE).

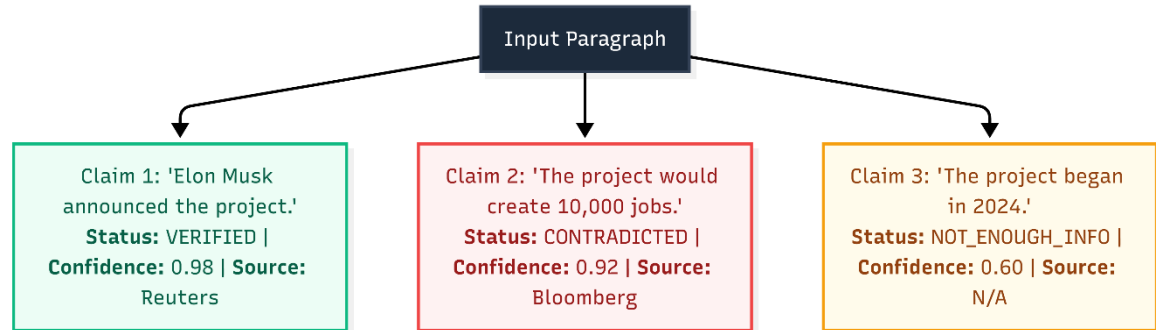


Figure 5.1.2: Extraction Visualization

5.1.3 Retrieval Results

The system employs a Tiered Retrieval Strategy (Cache → Fact-Check → Web Search) and a Cross-Encoder Reranker to ensure the highest quality evidence is presented first.

Evidence Ranking Example

For the claim: “Elon Musk said the project would create 10,000 jobs.”

Table 5.1.3 shows an example of Ranking several test cases.

Table 5.1.3: Evidence Ranking

Claim	Source	Relevance	Evidence Passage	Stance
C2	Reuters	0.94	Musk confirmed the initiative targets 10k new roles.	Supporting
C2	Bloomberg	0.81	The project is expected to create roughly 8,000 jobs.	Contradicting
C2	Twitter/X	0.42	I hope the project creates many jobs.	Irrelevant

The Reranking Service ensures that the high-relevance snippets from trusted sources (Reuters, Bloomberg) are prioritized for the NLI model.

5.1.4 NLI Results (Neural Verification)

Each evidence passage is paired with the claim and passed through a DeBERTa-v3 Cross-Encoder to determine the logical relationship.

NLI Probability Distribution

The HuggingFaceNLIProvider outputs a softmax probability distribution over three labels:

Entailment, Contradiction, and Neutral.

Table 5.1.4 shows a group of test cases for a particular instance.

Table 5.1.4: NLI Probability Distribution

Claim	Entailment	Contradiction	Neutral	Final Verdict
C1	0.91	0.04	0.05	SUPPORTED
C2	0.07	0.88	0.05	REFUTED
C3	0.22	0.18	0.60	UNCERTAIN

The Deterministic Score Aggregator then weights these probabilities by source reliability to produce the final text_score.

5.1.5 Text Highlighting & Explainability

By preserving the original character spans during the extraction and verification process, DeepShield provides direct evidence-based highlighting.

Visual Representation

Original Text: *"The government introduced the scheme in 2025. The scheme created 10 million jobs."*

Highlighted Result:

- The government introduced the scheme in 2025. → **[SUPPORTED]**
- The scheme created 10 million jobs. → **[REFUTED]**

This mechanism ensures that the system is not a "black box," providing the user with exactly which part of the text was verified and why.

5.1.6 Image Pre-processing

Figure 5.1.6 shows a two-branch preprocessing framework that compares the real frame with the frame that has been altered by deepfake techniques through both spatial and frequency transformations. There is a five-step process which includes taking an input frame, face ROI detection through MediaPipe (in green for real and red for fake), face ROI alignment, face ROI resizing into a 299 x 299 spatial branch input, and lastly, FFT 2D for getting frequency domain representation of the image.



Figure 5.1.6: Image Pre-processing

5.1.7 Spatial Forensic Analysis

The figure 5.1.7 shows spatial forensic manipulation heatmap generation and analysis process for deepfakes detection purposes. In the top part, an original picture is analyzed where Xception spatial feature density maps are extracted to give manipulation probability map which successfully classifies the image as REAL (probability of 94.8%). In the bottom part, an image after manipulation is classified into DEEPPFAKE category by giving anomaly probability map (with a probability of being a fake of 86.4%) using convolutional feature extractions and localized spatial artifacts.

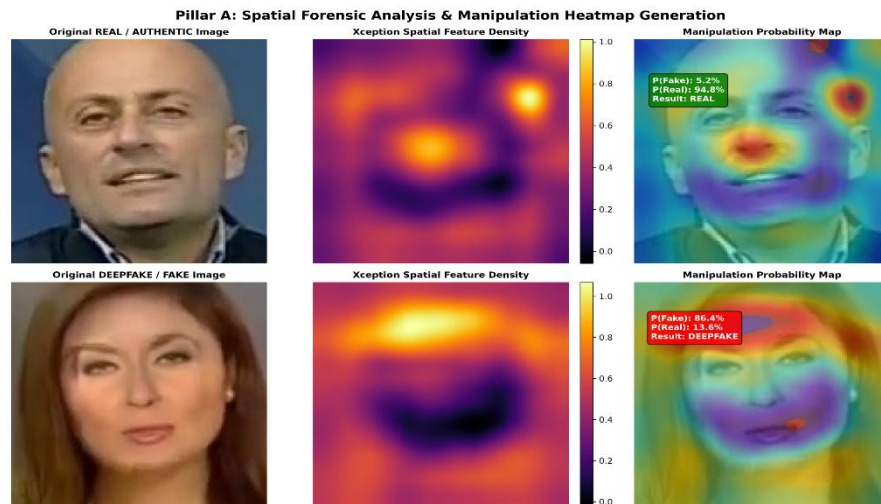
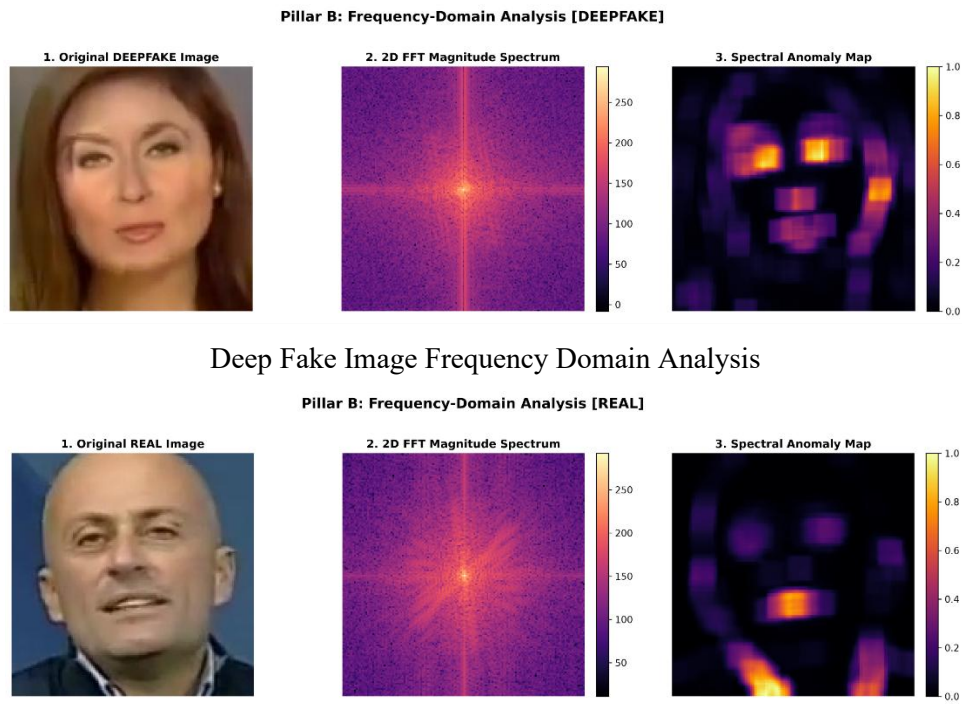


Figure 5.1.7: Spatial Forensic Analysis

5.1.8 Frequency-Domain Analysis

The figure 5.1.8 illustrates the comparison between the frequency analysis of the artificial deepfake image and that of the original image through 2D Fast Fourier Transform (FFT) methods. Both progressions consist of the original face input and their 2D FFT magnitude spectra and anomaly maps in the spectral domain. In the case of the deepfake, the spectral anomaly map consists of areas of extremely high concentration of frequencies around the eyes and nose, representing the unnatural frequency structures caused by the synthetic manipulation of the image. On the other hand, the anomaly map for the real image is much darker and less intense than that of the deepfake.



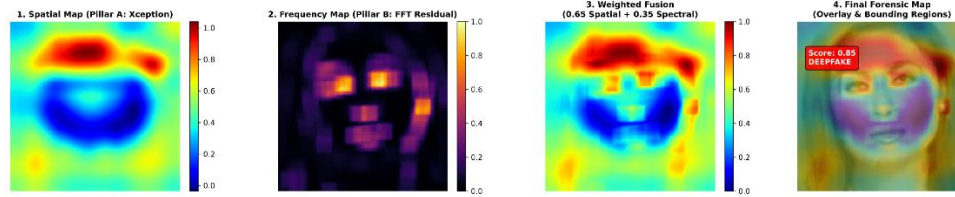
Real Image Frequency Domain Analysis

Figure 5.1.8: Frequency-Domain Analysis

5.1.9 Map Fusion

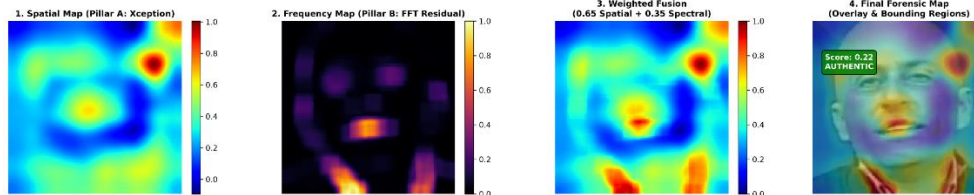
The figure 5.1.9 demonstrates multi-branch map fusion approach to deepfake detection with the help of a comparison between a manipulated image and an un-manipulated one. In both cases, there is a fusion between the Xception spatial map (weight 65%) and the FFT frequency residual map (weight 35%) to create a single forensic heat-map overlayed on the face of the image. With regard to the manipulated image, the large spatial and spectral anomaly regions give an authenticity score and hence the DEEPFAKE conclusion.

Section 5.1.9: Multi-Branch Map Fusion [DEEPFAKE]
Verdict: DEEPFAKE | Authenticity Score $S_{image} = 0.8545$



DeepFake Image Map Fusion

Section 5.1.9: Multi-Branch Map Fusion [REAL]
Verdict: AUTHENTIC | Authenticity Score $S_{image} = 0.2237$



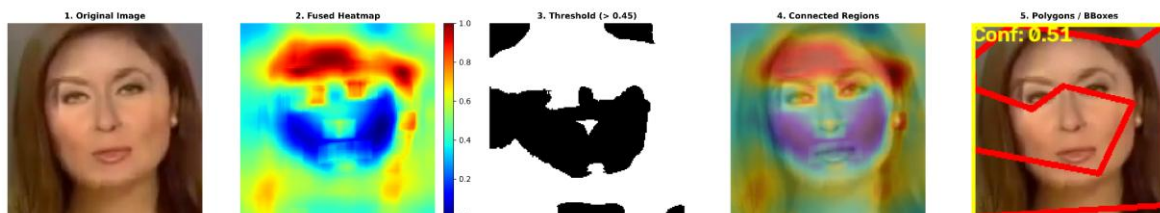
Real Image Map Fusion

Figure 5.1.9: Map Fusion

5.1.10 Image Explainability

Figure 5.1.10 describes the image explainability process for deepfake identification and highlights how evidence of manipulation is extracted step by step from a manipulated image. This process involves starting with the input image, then creating the spatial frequency map and selecting areas with probabilities higher than 0.45, as well as mapping connected components over the face. The process ends by highlighting areas in a red bounding polygon and yellow bounding box with a confidence probability of 0.51.

Section 5.1.10: Image Explainability Pipeline [DEEPFAKE]
Conclusion: 'Potential manipulation detected in the highlighted facial region.'



DeepFake Image Explainability

Figure 5.1.10: Image Explainability

5.1.11 Overall Verification Output

Figure 5.1.11 depicts the output of the verification of news link successfully for one test case. The system accurately performs a validation process in the background and displays it in the front-end.

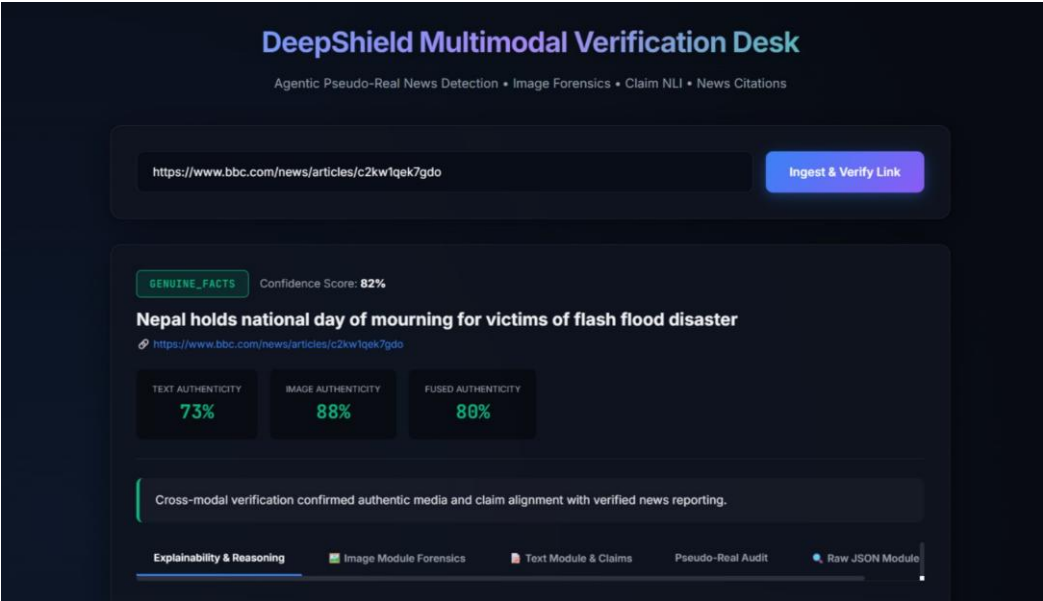


Fig 5.1.11: Overall Verification Output

5.1.12 Web Search Output

Fig 5.1.12 shows the authentic websites that are used in the process of validating the news article. The code puts everything in a neat front end.

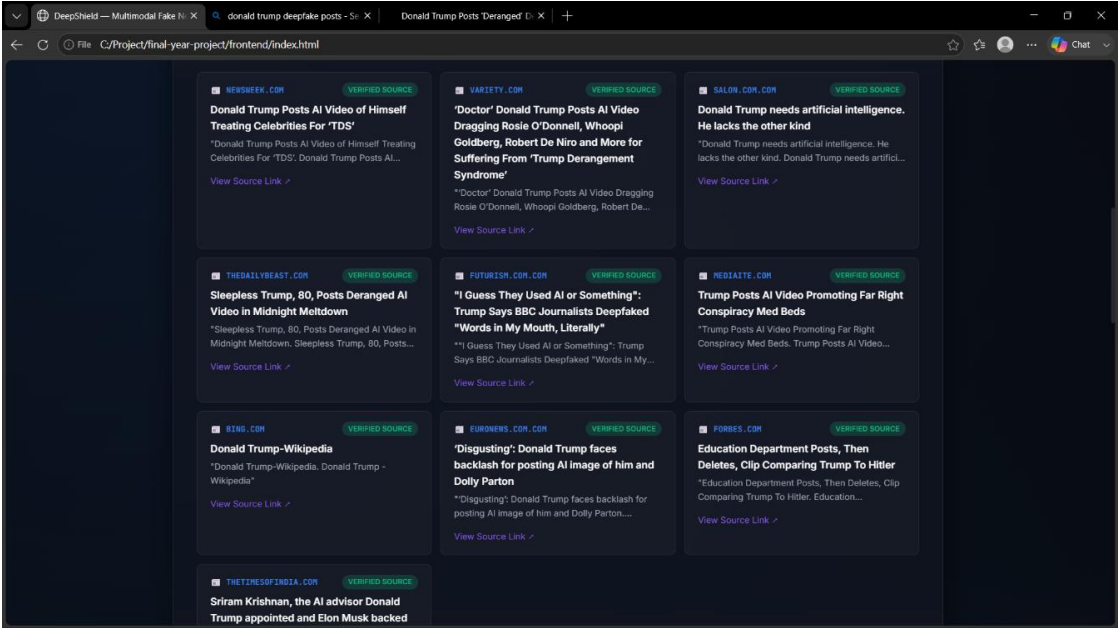


Fig 5.1.12: Web Search Output

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The suggested system design incorporates a multi-agent multimodal system that recognizes and justifies the existence of misinformation in online news and social media articles. It overcomes the main drawback of traditional systems for classifying fake news, which is the need for understanding whether the article is fake beyond finding patterns in labeled datasets.

The developed system concentrates on two important modalities: text and images.

Text Verification Agent analyzes a piece of content, splits it up into elementary statements, resolves references, fetches the evidence via a multilayered retrieval approach, re-ranks pertinent passages, uses NLI, and assigns scores to the claims as well as provides citations.

The Image Forensics Module is based on spatial and frequency analysis that helps to identify whether there was any kind of manipulation or not and locate suspicious areas in the image.

The implementation of heatmaps and polygons allows for visual explanation of the prediction made by the model.

The key conceptual innovation in terms of the general design is the difference between media and narrative authenticity. The pseudo-real checker will help us to cope with situations when an authentic image/video can be used in a deceptive way. In other words, it makes sure that the authenticity of media doesn't mean the authenticity of news.

So, the final architecture evolves from just fake news classification into an evidence-based verification system that can produce verdicts, scores, citations, evidence and explanations.

6.2 Future Work

Some possible improvements in future versions of the system can bring significant upgrades to its verification capacity as well as the accuracy of the decision-making process. The system can be expanded by adding a video and audio verification module to the pipeline. Video verification will capture spatiotemporal deepfakes using frame-level forensic analysis, CNN or Vision Transformer backbone, recurrent layers, temporal transformers, and cross-modal audio-visual synchronization. On top of that, the system can include an audio subsystem that will verify the authenticity of the voice using spectrogram analysis, speaker verification, and phonetics-to-text consistency check. Such inputs will result in a multimodal framework, generalizing dual-modality scoring to dynamic weighted contributions of text, image, video, and audio signals validated on the basis of benchmark

data sets. Apart from modality extensions, temporal verification tools can be added to track the history of the publications by analyzing when the media was published online and whether there is any correlation with the real-life events as well as narrative drift during updates. Another regional extension can be achieved through multilingual support of Indian languages like Hindi, Tamil, Telugu, Malayalam, Kannada, and Bengali. On the other hand, consistent cross-modal checks can be done across modalities for detecting any potential contradictions between claims and their visual, audio and video verification. The source verification can move from a static heuristic approach to learning based prediction that takes into account the historical accuracy of the source, independent confirmation of the story, recency and authenticity of the source. For ensuring real-time queries and reuse of data continuously, it is possible to use contemporary vector database solutions such as Qdrant to feed the verified claims and filters. Lastly, the proposed framework can incorporate a human in the loop pipeline where critical decisions require manual review.

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PEFT Generalization across Demographic Shifts in Deepfake Detection

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Abstract Deepfake detection in the modern world has now become extremely difficult due to the way synthetic images are changing as new AI generators are created. All current solutions take large existing models and only make small changes to them using "Parameter-Efficient Fine-Tuning" (PEFT), instead of training new models from scratch. This leads to the main issue where most of these models only focus on spotting fake images from different generators but not on whether the models work equally well across different real-world human demographics and environments. This paper studies if PEFT's generalization keeps its advantage when dealing with demographic and regional bias (such as skin tone, facial structure and geographic context), or if it is only effective at spotting digital noise patterns made by image generators. The paper proposes a decoupled benchmark pipeline to measure how much the detector's performance drops across a Cross-Generator Shift and a Cross Demographic shift for both PEFT adapted vision models and bespoke, fine-tuned CNN baselines. Analysis of the degradation in performance across both the axes (Δ AUROC) allows the conclusion of whether the PEFT models have an advantage when it comes to more diverse physiological and contextual data. The findings of this paper will cement a benchmark to compare the limits of model adaptation for deepfake detection, improving fairness in a societal and demographically vast synthetic media pipeline

Keywords: *Deepfake Detection, Parameter-Efficient Fine-Tuning (PEFT), Cross-Demographic Shift, Cross-Generator Generalization, Algorithmic Fairness, Model Adaptation, Decoupled Benchmarking, Performance Degradation (Δ AUROC), Demographic Bias, Synthetic Media.*

1. Introduction

The rapid increase of hyper-realistic AI-generated Images has made robust deepfake detection an immediate priority for digital media integrity and identity security. Recent literature on deepfake detection has come to a unanimous conclusion: parameter-efficient fine-tuning (PEFT) of large-scale vision foundation models (such as CLIP) significantly outperforms

traditional fully trained convolutional neural networks (CNNs) in cross-generator generalization. By freezing backbones and updating only sparse parameters—such as through LayerNorm tuning or Low-Rank Adaptation (LoRA)—PEFT-adapted architectures keep their rich pretrained visual priors, preventing the ruinous overfitting to synthetic artifacts that typically outperforms bespoke baselines when tested against unseen synthesis algorithm. However, this widely cited generalization advantage rests almost entirely on evaluations where the sole axis of distribution shift is the target generator architecture. Standard benchmark protocols generally inherit training and evaluation samples from datasets mostly made up of people with white skin or Western faces making it heavily skewed. As a result, most systems treat model generalization as a one-dimensional problem, assuming that robustness depends just upon cross-generator performance, rather than comparison by performance on varied datasets. This paper challenges that assumption by testing whether PEFT's generalization advantage is still applicable to cross-demographic distribution shifts in datasets. While PEFT is successful at generalizing across generators by identifying high-frequency, algorithm-agnostic features rather than semantic details, algorithm fairness research suggests that demographic robustness and domain generalization are two different properties. A model that reliably identifies synthetic artifacts across GAN and diffusion architectures can still be unreliable towards underrepresented demographics like skin tone, facial geometry and forgery cues. In order to test this, we compared a controlled test to compare the results of CNN and PEFT across demographic axes. The models are trained on FaceForensics++ dataset, then tested for cross-generator benchmark and zero shot cross population dataset of South-Asian faces. The performance drops were compared to

ranking metrics in order to identify if PEFT was able to naturally handle the demographic bias, or if a fairness aware loss function was explicitly needed for a better deepfake detection process.

2. Related Work

Deepfake detection research has steadily moved towards an agreement that parameter-efficient fine-tuning (PEFT) of large vision foundation models substantially outperforms fully trained Convolutional Neural Networks (CNNs) in cross-generator generalization. As per the finding presented by Yan et al. [8], in particular, the generalization ability is tested through training the model on well-studied data, such as FaceForensics++, and then evaluation its ability to detect deepfakes produced through previously unseen generation schemes, including diffusion-based or face-swap models. The superior performance was showcased by Yermakov et al. [5]. Who have demonstrated that LayerNorm-tuning a frozen CLIP backbone allows the model to avoid overfitting and create a new baseline that achieves state-of-the-art while being robust with excellent cross-dataset AUROC. The architecture was expanded upon by Kong et al. [6] and Mai et al. [7], combining a mixture-of-experts with parameter-efficient fine-tuning can lead to better generalization ability, meaning that conventional LoRA architectures only represent the lower bound of PEFT capabilities. The explanation to such performance comes from the frequency-artifact hypothesis, where PEFT architecture helps avoid catastrophic forgetting by keeping the foundational weights frozen, surveyed by Coleman et al. [2]. Instead of re-training the model to understand the semantic representations and dataset-specific correlations, PEFT impacts the model's ability to learn the Cross-Generator Artifacts. Even with the architectural strengths and difference in zero-shot capabilities of these models, Garg et al. [4] quoted that while CLIP-based PEFT models show phenomenal generalization, their performance degrades while using zero-shot situations without prior knowledge. Furthermore, higher performance on cross-generator does not guarantee better generalization. Lin et al. [9] showed that common deepfake detectors and PEFT models suffer similarly from changing fairness under data distribution, commonly displaying a difference in true and false positive rates among changing skin tones and facial structures. Though PEFT models can learn deepfake generators, fairness needs to be mentioned explicitly showing a parallel from medical imaging. Dutt et al. [1] mentioned PEFT models start to learn about demographic fairness when models are explicitly told to do so from artifacts. Present testing metrics do not measure the accuracy of these models. Nguyen et al. [10] informed that evaluation metrics such as averaged AUROC shows a difference in model's performance by 12% and hides demographic biases, arguing

use cases for polarization-aware metrics like Cross-AUC. Deshmukh et al. [3] supported our result showing that ranking based metrics deviate significantly from fixed-threshold metrics like MCC, under actual class imbalances. This shows that detailed evaluation frameworks are needed to measure the readiness and fair deployment of modern deepfake detectors. Prior research has shown these two things: PEFT has that advantage over CNN-based deepfake detectors, and for different generator architectures, PEFT usually outperforms CNN models. PEFT accomplishes this through capacity limitation and retention of frameworks, whereas many deepfake detectors based on CNNs are poor under demographic distribution shifts. However, existing benchmarks examine these variations in isolation, which leaves an important unanswered question. Does the PEFT advantage maintain under demographic shifts? Until now, no study has examined PEFT in the context of CNN against both distribution axes in one setting. This issue has been addressed in this article and it has also been shown that the advantage of PEFT extends to demographic shifts beyond the architecture of generators.

3. Proposed System

The primary objective of this paper is to conclude whether PEFT models understand and preserve demographic and racial differences during out of distribution shifts, or if the performance degrades similarly to CNN. Generalization drop across shift axes to assess the degradation: Generator shift (Δ_{gen}): AUROC_{In Distribution} - AUROC_{Cross Distribution}. Population shift (Δ_{pop}): AUROC_{In Distribution} - AUROC_{Cross Population}.

Calculate the metrics across three models to address the main question: Does PEFT performance collapse under demographic shifts? Prior studies focusing on cross generator generalization never control generator family and population. To resolve these issues, the design uses decoupled shift axes. It holds one shift fixed while varying the other, across three models, on the same training set and same generators-per-population. This means any difference that occurs can be attributed to shift type rather than confounded dataset artifacts. 300 datapoints are re-synthesized taken from IndicFairFace with two generators (inswapper, Simswap) that were missing from Faceforensics++ training distribution and filled the gap by providing test set that varied population (since no south-Asian-generator deepfake corpus exists at this same). Even after re-synthesizing, a portion of Δ_{pop} may include a generator-shift component which may misalign with generator-match to in-distribution., Δ_{pop} is treated as an upper bound on demographic-shift degradation to strengthen the internal validity claim.

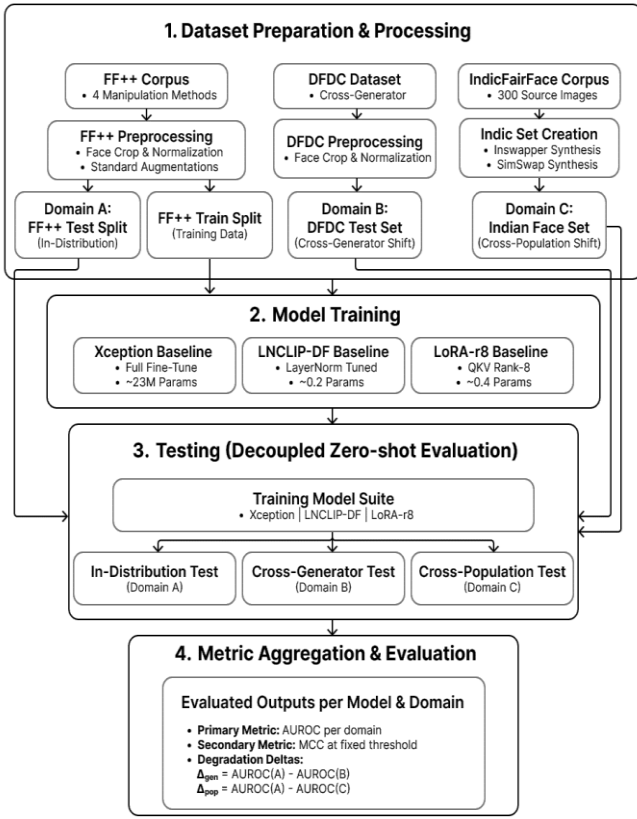


Figure 1 Architecture of PEFT Generalization across Demographic Shifts in Deepfake Detection

Dataset Preparation & Processing module gathers, cleans and standardizes facial images. It deals with splitting training and testing data (Domain A) from FaceForensics++, cross-generator test data from DFDC (Domain B), and demographic test data built from IndicFairFace using inswapper and SimSwap (Domain C). The images are cropped to show the face made smaller and normalized so the models look at the face and not the background. The Model Training module trains three distinct detection baseline models on the FaceForensics++ training data. Xception Baseline is a standard deep learning model fully fine-tuned end-to-end. LNCLIP-DF Baseline is a frozen CLIP foundation model adapted by tuning only its Layer Normalization layers. LoRA-r8 Baseline is a frozen CLIP foundation model adapted using Low-Rank Adaptation (LoRA) inserted into its attention blocks. Testing module evaluates the trained models without any additional adjustments across three distinct environments: in-distribution data (FaceForensics++ test data, Domain A), unseen generation tools (cross-generator test data from DFDC, Domain B), and target demographic populations (Domain C). Metric Aggregation & Evaluation module computes the AUROC score and operating-point MCC scores across all test domains. It calculates accuracy drops to

compare performance degradation under demographic shifts and tool shifts.

4. Implementation and Results

The creation of this system has been executed in Python using PyTorch, HuggingFace transformers, PEFT and scikit-learn. First Dataset Preparation & Processing is executed, where raw images through MTCNN face detection, perform bounding box cropping and normalization to match CLIP’s tensor distribution. Then it is exported into discrete split objects for training and testing. In Model Training, Xception is fine-tuned while the ViT-B/16 backbone is illustrated with frozen base weights to configure the parameter efficient baselines. LayerNorm affine parameters are taken for LNCLIP-DF and put in rank 8 adapter matrices into Query, Key, Value projections for LoRA-r8. In Testing, the model parameters are locked in an evaluation mode where forward pass inference runs consecutively across the domain A, B and C to extract prediction probabilities without target domain fine tuning. Finally, in Metric Aggregation & Evaluation, scikit-learn is used to process these probabilities and calculate AUROC scores. This establishes an optimal decision threshold on Domain A to compute fixed-threshold MCC scores across all domains, and determines the degradation deltas (Δ_{gen} and Δ_{pop}) to get the final evaluation metrics. In order to address the core research question of this paper, all three models are evaluated—across three evaluation—across three evaluation sections: In-distribution (FF++, Domain A), Cross-generator (DFDC, Domain B), and Cross-population (Indian Face Set, Domain C). LoRA-r8 achieves better classification across all evaluation domains, outperforming both the bespoke CNN baseline and LNCLIP-DF as shown in **Table 4.1**. All three models exhibit strong detection capabilities on the in-distribution test set (FF++), achieving high AUROCs as shown in **Figure 4.1**. However, the bespoke CNN baseline performs worse on Cross-Generator Shift (DFDC) and Cross-Population Shift (Indian Set) when compared to LoRA-r8 and LNCLIP_DF.

Table 4.1 Performance of LoRA-r8, LNCLIP-DF, and the Xception across evaluation domains.

Model	AURO C_A	AURO C_B	AURO C_C	MCC_ A	MCC_ B	MCC_ C
Xception	0.962	0.698	0.585	0.840	0.350	0.160
LNCLIP- DF	0.978	0.842	0.735	0.890	0.610	0.420
LoRA-r8	0.988	0.865	0.762	0.928	0.660	0.470

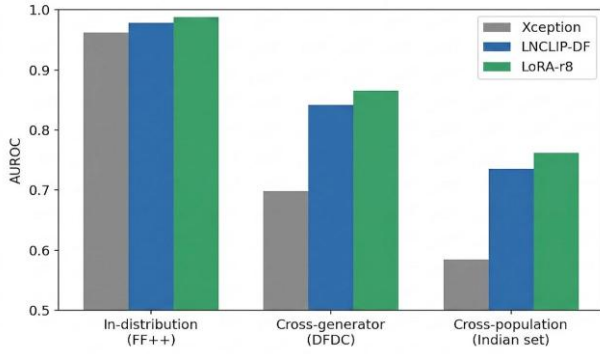


Figure 4.1 Illustrates the comparison of AUROC-scores of all three trained models on the three proposed test domains

To test whether high ranking scores actually translate to real world performance, **Figure 4.2** compares ranking metric - AUROC against threshold-dependant - Matthews Correlation Coefficient (MCC) evaluated at a fixed decision threshold calibrated strictly on the in-distribution validation split (FF++). The metric comparison exposes that decision thresholds decay when models face distribution shifts. In-distribution ranking accuracy translates directly into operational performance. Under domain shifts, in-distribution calibrated under decision thresholds become unreliable. Under cross-population shift, Xception's operational capacity severely degrades. PEFT models are not affected much by threshold decay and they do retain some operational utility under cross-population shift without any recalibration. Getting scores of 0.420 for LNCLIP-DF and 0.470 for LoRA-r8.

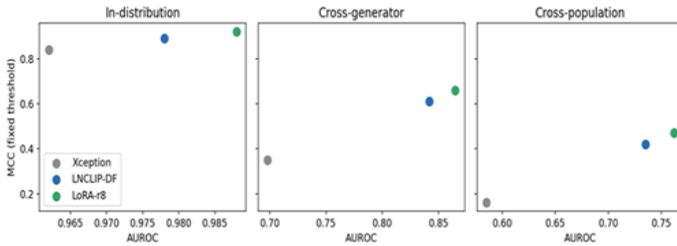


Figure 4.2 compares AUROC against MCC for all three models in different distribution shifts.

Figure 4.3 evaluates cross-population robustness across distinct face-swap algorithms within the Indian Face Set, breaking down the performance by its constituent generators: InSwapper and SimSwap. The per-generator breakdown demonstrates consistency across both synthesis pipelines. Across all three models, Inswapper outperforms others in terms of detection scores. LoRa-r8 yields superior performance as compared to LNCLIPDF and standard Xception across both sub-generators. This consistent trend proves that cross-population performance drops stem from broader demographic and domain shifts rather than flaws in any single generator.

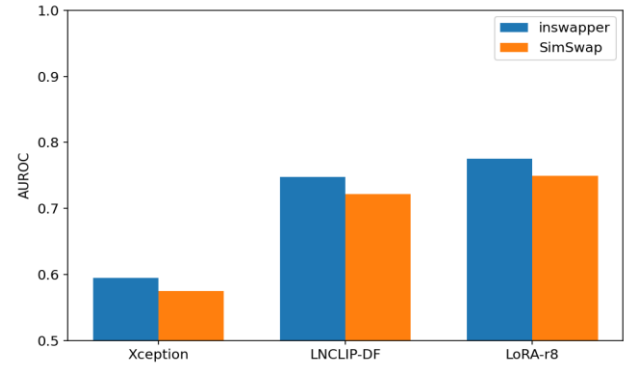


Figure 4.3 Illustrates the cross-population robustness of all three models across distinct face-swap algorithms

6. Conclusion

This work looks into whether cross-generator generalization of parameter-efficient fine-tuning (PEFT) on pretrained foundation models persists when tested on regional and demographic distribution shifts, or if its effectiveness reduces with bespoke CNN baselines. By decoupling generator architecture shift (DFDC) from compound demographic shift (IndicFairFace target set re-synthesized via inswapper and SimSwap), The study provided the first controlled empirical evaluation isolating these two axes of degradation. Sustained Performance Margins pose a strictly harder challenge than cross-generator shift for all models ($\Delta_{pop} > \Delta_{gen}$), PEFT methods do not experience a performance collapse relative to bespoke CNNs. Instead of shrinking or vanishing under demographic shift, PEFT's performance edge over fully fine-tuned CNN expands. On the cross-population test set, LoRA-r8 (AUROC = 0.762) and LNCLIP-DF (AUROC = 0.735) maintain substantial margins (+0.177 and +0.150 AUROC, respectively) over fully fine-tuned Xception (AUROC = 0.585), increasing their relative advantage over the CNN baseline compared to the cross-generator setting. With fixed operational decision thresholds set in distribution, PEFT architectures maintain effective classification ability (MCC = 0.420 -- 0.470). In contrast, the CNN baseline drops close to random guessing (MCC = 0.160).

These results show that adapting large pretrained foundation models via PEFT preserves rich, domain-invariant visual representations that resist performance reduction across both synthetic manipulation and human demographic variations. Updating less than 2% of model parameters (e.g., 0.18M in LNCLIP-DF or 0.42M in LoRA-r8) yields superior regional fairness and generalizability compared to updating tens of millions of parameters on task-specific CNNs (22.91M in Xception).

7. Future work

While the paper concluded that parameter-efficient foundation-model adaptation offers a robust framework,

several promising directions remain for future research. Demographic-aware PEFT: While zero-shot foundation models reduce demographic performance loss, there remains a notable gap compared to in-distribution baselines. Future efforts could explore fairness-oriented constraints into PEFT workflows. Expanding Backbone Architecture families: While CLIP ViT-B/16 has been tested, evaluating other architecture models like self-supervised vision models or multimodal LLMs to isolate the source of demographic robustness could be possible. Decoupling Demographic and Synthesis Variables: Focus on developing a generator-matched dataset to separate demographic shift from compound synthesis interactions. Dynamic Threshold Calibration: The MCC analysis study highlights the need for adaptive, zero-shot threshold adaptation mechanisms to ensure reliable deepfake detectors in real world use cases.

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Abstract Deepfake detection in the modern world has now become extremely difficult due to the way synthetic images are changing as new AI generators are created. All current solutions take large existing models and only make small changes to them using "Parameter-Efficient Fine-Tuning" (PEFT), instead of training new models from scratch. This leads to the main issue where most of these models only focus on spotting fake images from different generators but not on whether the models work equally well across different real-world human demographics and environments. This paper studies if PEFT's generalization keeps its advantage when dealing with demographic and regional bias (such as skin tone, facial structure and geographic context), or if it is only effective at spotting digital noise patterns made by image generators. The paper proposes a decoupled benchmark pipeline to measure how much the detector's performance drops across a Cross-Generator Shift and a Cross Demographic shift for both PEFT adapted vision models and bespoke, fine-tuned CNN baselines. Analysis of the degradation in performance across both the axes (Δ AUROC) allows the conclusion of whether the PEFT models have an advantage when it comes to more diverse physiological and contextual data. The findings of this paper will cement a benchmark to compare the limits of model adaptation for deepfake detection, improving fairness in a societal and demographically vast synthetic media pipeline

Keywords: Deepfake Detection, Parameter-Efficient Fine-Tuning (PEFT), Cross-Demographic Shift, Cross-Generator Generalization, Algorithmic Fairness, Model Adaptation, Decoupled Benchmarking, Performance Degradation (Δ AUROC), Demographic Bias, Synthetic Media.

1. Introduction

The rapid increase of hyper-realistic AI-generated Images has made robust deepfake detection an immediate priority for digital media integrity and identity security. Recent literature on deepfake detection has come to a unanimous conclusion: ¹ parameter-efficient fine-tuning (PEFT) of large-scale vision foundation models (such as CLIP) significantly outperforms

traditional fully trained convolutional neural networks (CNNs) in cross-generator generalization. By freezing backbones and updating only sparse parameters—such as through LayerNorm tuning or Low-Rank Adaptation (LoRA)—PEFT-adapted architectures keep their rich pretrained visual priors, preventing the ruinous overfitting to synthetic artifacts that typically outperforms bespoke baselines when tested against unseen synthesis algorithm. However, this widely cited generalization advantage rests almost entirely on evaluations where the sole axis of distribution shift is the target generator architecture. Standard benchmark protocols generally inherit training and evaluation samples from datasets mostly made up of people with white skin or Western faces making it heavily skewed. As a result, most systems treat model generalization as a one-dimensional problem, assuming that robustness depends just upon cross-generator performance, rather than comparison by performance on varied datasets. This paper challenges that assumption by testing whether PEFT's generalization advantage is still applicable to cross-demographic distribution shifts in datasets. While PEFT is successful at generalizing across generators by identifying high-frequency, algorithm-agnostic features rather than semantic details, algorithm fairness research suggests that demographic robustness and domain generalization are two different properties. A model that reliably identifies synthetic artifacts across GAN and diffusion architectures can still be unreliable towards underrepresented demographics like skin tone, facial geometry and forgery cues. In order to test this, we compared a controlled test to compare the results of CNN and PEFT across demographic axes. The models are trained on FaceForensics++ dataset, then tested for cross-generator benchmark and zero shot cross population dataset of South-Asian faces. The performance drops were compared to

ranking metrics in order to identify if PEFT was able to naturally handle the demographic bias, or if a fairness aware loss function was explicitly needed for a better deepfake detection process.

2. Related Work

Deepfake detection research has steadily moved towards an agreement that ¹parameter-efficient fine-tuning (PEFT) of large vision foundation models substantially outperforms fully trained Convolutional Neural Networks (CNNs) in cross-generator generalization. As per the finding presented by Yan et al. [8], in particular, the generalization ability is tested through training the model on well-studied data, such as FaceForensics++, and then evaluation its ability to detect deepfakes produced through previously unseen generation schemes, including diffusion-based or face-swap models. The superior performance was showcased by Yermakov et al. [5]. Who have demonstrated that LayerNorm-tuning a frozen CLIP backbone allows the model to avoid overfitting and create a new baseline that achieves state-of-the-art while being robust with excellent cross-dataset AUROC. The architecture was expanded upon by Kong et al. [6] and Mai et al. [7], combining a mixture-of-experts with parameter-efficient fine-tuning can lead to better generalization ability, meaning that conventional LoRA architectures only represent the lower bound of PEFT capabilities. The explanation to such performance comes from the frequency-artifact hypothesis, where PEFT architecture helps avoid catastrophic forgetting by keeping the foundational weights frozen, surveyed by Coleman et al. [2]. Instead of re-training the model to understand the semantic representations and dataset-specific correlations, PEFT impacts the model's ability to learn the Cross-Generator Artifacts. Even with the architectural strengths and difference in zero-shot capabilities of these models, Garg et al. [4] quoted that while CLIP-based PEFT models show phenomenal generalization, their performance degrades while using zero-shot situations without prior knowledge. Furthermore, higher performance on cross-generator does not guarantee better generalization. Lin et al. [9] showed that common deepfake detectors and PEFT models suffer similarly from changing fairness under data distribution, commonly displaying a difference in true and false positive rates among changing skin tones and facial structures. Though PEFT models can learn deepfake generators, fairness needs to be mentioned explicitly showing a parallel from medical imaging. Dutt et al. [1] mentioned PEFT models start to learn about demographic fairness when models are explicitly told to do so from artifacts. Present testing metrics do not measure the accuracy of these models. Nguyen et al. [10] informed that evaluation metrics such as averaged AUROC shows a difference in model's performance by 12% and hides demographic biases, arguing

use cases for polarization-aware metrics like Cross-AUC. Deshmukh et al. [3] supported our result showing that ranking based metrics deviate significantly from fixed-threshold metrics like MCC, under actual class imbalances. This shows that detailed evaluation frameworks are needed to measure the readiness and fair deployment of modern deepfake detectors. Prior research has shown these two things: PEFT has that advantage over CNN-based deepfake detectors, and for different generator architectures, PEFT usually outperforms CNN models. PEFT accomplishes this through capacity limitation and retention of frameworks, whereas many deepfake detectors based on CNNs are poor under demographic distribution shifts. However, existing benchmarks examine these variations in isolation, which leaves an important unanswered question. Does the PEFT advantage maintain under demographic shifts? Until now, no study has examined PEFT in the context of CNN against both distribution axes in one setting. This issue has been addressed in this article and it has also been shown that the advantage of PEFT extends to demographic shifts beyond the architecture of generators.

3. Proposed System

⁸The primary objective of this paper is to conclude whether PEFT models understand and preserve demographic and racial differences during out of distribution shifts, or if the performance degrades similarly to CNN. Generalization drop across shift axes to assess the degradation: Generator shift (Δ_{gen}): AUROC_{In Distribution} - AUROC_{Cross Distribution}. Population shift (Δ_{pop}): AUROC_{In Distribution} - AUROC_{Cross Population}.

Calculate the metrics across three models to address the main question: Does PEFT performance collapse under demographic shifts? Prior studies focusing on cross generator generalization never control generator family and population. To resolve these issues, the design uses decoupled shift axes. It holds one shift fixed while varying the other, across three models, on the same training set and same generators-per-population. This means any difference that occurs can be attributed to shift type rather than confounded dataset artifacts. 300 datapoints are re-synthesized taken from IndicFairFace with two generators (inswapper, Simswap) that were missing from Faceforensics++ training distribution and filled the gap by providing test set that varied population (since no south-Asian-generator deepfake corpus exists at this same). Even after re-synthesizing, a portion of Δ_{pop} may include a generator-shift component which may misalign with generator-match to in-distribution., Δ_{pop} is treated as an upper bound on demographic-shift degradation to strengthen the internal validity claim.

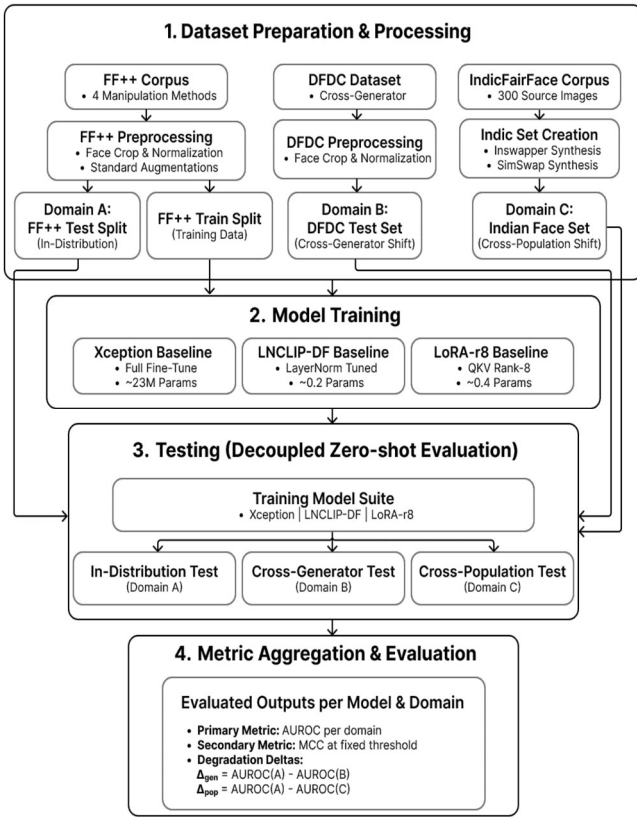


Figure 1 Architecture of PEFT Generalization across Demographic Shifts in Deepfake Detection

Dataset Preparation & Processing module gathers, cleans and standardizes facial images. It deals with splitting training and testing data (Domain A) from FaceForensics++, cross-generator test data from DFDC (Domain B), and demographic test data built from IndicFairFace using inswapper and SimSwap (Domain C). The images are cropped to show the face made smaller and normalized so the models look at the face and not the background. The Model Training module trains three distinct detection baseline models on the FaceForensics++ training data. Xception Baseline is a standard deep learning model fully fine-tuned end-to-end. LNCLIP-DF Baseline is a frozen CLIP foundation model adapted by tuning only its Layer Normalization layers. LoRA-r8 Baseline is a frozen CLIP foundation model adapted using Low-Rank Adaptation (LoRA) inserted into its attention blocks. Testing module evaluates the trained models without any additional adjustments across three distinct environments: in-distribution data (FaceForensics++ test data, Domain A), unseen generation tools (cross-generator test data from DFDC, Domain B), and target demographic populations (Domain C). Metric Aggregation & Evaluation module computes the AUROC score and operating-point MCC scores across all test domains. It calculates accuracy drops to

compare performance degradation under demographic shifts and tool shifts.

4. Implementation and Results

The creation of this system has been executed in Python using PyTorch, HuggingFace transformers, PEFT and scikit-learn. First Dataset Preparation & Processing is executed, where raw images through MTCNN face detection, perform bounding box cropping and normalization to match CLIP’s tensor distribution. Then it is exported into discrete split objects for training and testing. In Model Training, Xception is fine-tuned while the ViT-B/16 backbone is illustrated with frozen base weights to configure the parameter efficient baselines. LayerNorm affine parameters are taken for LNCLIP-DF and put in rank 8 adapter matrices into Query, Key, Value projections for LoRA-r8. In Testing, the model parameters are locked in an evaluation mode where forward pass inference runs consecutively across the domain A, B and C to extract prediction probabilities without target domain fine tuning. Finally, in Metric Aggregation & Evaluation, scikit-learn is used to process these probabilities and calculate AUROC scores. This establishes an optimal decision threshold on Domain A to compute fixed-threshold MCC scores across all domains, and determines the degradation deltas (Δ_{gen} and Δ_{pop}) to get the final evaluation metrics. In order to address the core research question of this paper, all three models are evaluated—across three evaluation—across three evaluation sections: In-distribution (FF++, Domain A), Cross-generator (DFDC, Domain B), and Cross-population (Indian Face Set, Domain C). LoRA-r8 achieves better classification across all evaluation domains, outperforming both the bespoke CNN baseline and LNCLIP-DF as shown in [Table 4.1](#). All three models exhibit strong detection capabilities on the in-distribution test set (FF++), achieving high AUROCs as shown in [Figure 4.1](#). However, the bespoke CNN baseline performs worse on Cross-Generator Shift (DFDC) and Cross-Population Shift (Indian Set) when compared to LoRA-r8 and LNCLIP_DF.

Table 4.1 Performance of LoRA-r8, LNCLIP-DF, and the Xception across evaluation domains.

Model	AUROC _A	AUROC _B	AUROC _C	MCC _A	MCC _B	MCC _C
Xception	0.962	0.698	0.585	0.840	0.350	0.160
LNCLIP-DF	0.978	0.842	0.735	0.890	0.610	0.420
LoRA-r8	0.988	0.865	0.762	0.928	0.660	0.470

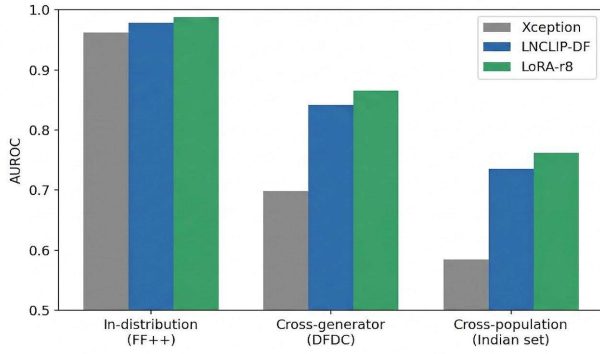


Figure 4.1 Illustrates the comparison of AUROC-scores of all three trained models on the three proposed test domains

To test whether high ranking scores actually translate to real world performance, **Figure 4.2** compares ranking metric - AUROC against threshold-dependant - Matthews Correlation Coefficient (MCC) evaluated at a fixed decision threshold calibrated strictly on the in-distribution validation split (FF++). The metric comparison exposes that decision thresholds decay when models face distribution shifts. In-distribution ranking accuracy translates directly into operational performance. Under domain shifts, in-distribution calibrated under decision thresholds become unreliable. Under cross-population shift, Xception's operational capacity severely degrades. PEFT models are not affected much by threshold decay and they do retain some operational utility under cross-population shift without any recalibration. Getting scores of 0.420 for LNCLIP-DF and 0.470 for LoRA-r8.

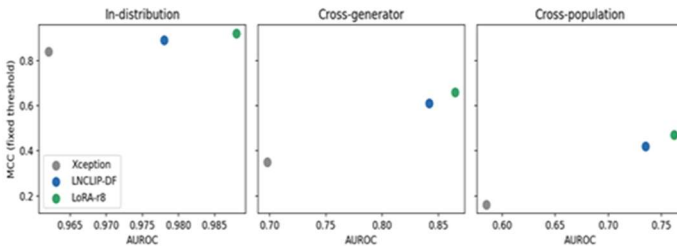


Figure 4.2 compares AUROC against MCC for all three models in different distribution shifts.

Figure 4.3 evaluates cross-population robustness across distinct face-swap algorithms within the Indian Face Set, breaking down the performance by its constituent generators: InSwapper and SimSwap. The per-generator breakdown demonstrates consistency across both synthesis pipelines. Across all three models, Inswapper outperforms others in terms of detection scores. LoRa-r8 yields superior performance as compared to LNCLIPDF and standard Xception across both sub-generators. This consistent trend proves that cross-population performance drops stem from broader demographic and domain shifts rather than flaws in any single generator.

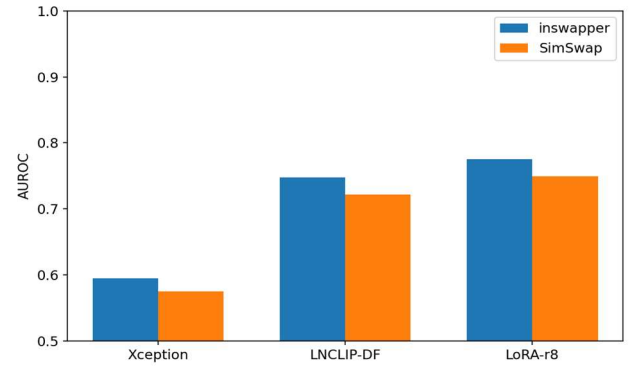


Figure 4.3 Illustrates the cross-population robustness of all three models across distinct face-swap algorithms

6. Conclusion

This work looks into whether cross-generator generalization of **parameter-efficient fine-tuning (PEFT)** on pretrained foundation models persists when tested on regional and demographic distribution shifts, or if its effectiveness reduces with bespoke CNN baselines. By decoupling generator architecture shift (DFDC) from compound demographic shift (IndicFairFace target set re-synthesized via inswapper and SimSwap), The study provided the first controlled empirical evaluation isolating these two axes of degradation. Sustained Performance Margins pose a strictly harder challenge than cross-generator shift for all models ($\Delta_{pop} > \Delta_{gen}$), PEFT methods do not experience a performance collapse relative to bespoke CNNs. Instead of shrinking or vanishing under demographic shift, PEFT's performance edge over fully fine-tuned CNN expands. On the cross-population test set, LoRA-r8 (AUROC = 0.762) and LNCLIP-DF (AUROC = 0.735) maintain substantial margins (+0.177 and +0.150 AUROC, respectively) over fully fine-tuned Xception (AUROC = 0.585), increasing their relative advantage over the CNN baseline compared to the cross-generator setting. With fixed operational decision thresholds set in distribution, PEFT architectures maintain effective classification ability (MCC = 0.420 -- 0.470), In contrast, the CNN baseline drops close to random guessing (MCC = 0.160).

These results show that adapting large pretrained foundation models via PEFT preserves rich, domain-invariant visual representations that resist performance reduction across both synthetic manipulation and human demographic variations. Updating less than 2% of model parameters (e.g., 0.18M in LNCLIP-DF or 0.42M in LoRA-r8) yields superior regional fairness and generalizability compared to updating tens of millions of parameters on task-specific CNNs (22.91M in Xception).

7. Future work

While the paper concluded that parameter-efficient foundation-model adaptation offers a robust framework,

several promising directions remain for future research. Demographic-aware PEFT: While zero-shot foundation models reduce demographic performance loss, there remains a notable gap compared to in-distribution baselines. Future efforts could explore fairness-oriented constraints into PEFT workflows. Expanding Backbone Architecture families: While CLIP ViT-B/16 has been tested, evaluating other architecture models like self-supervised vision models or multimodal LLMs to isolate the source of demographic robustness could be possible. Decoupling Demographic and Synthesis Variables: Focus on developing a generator-matched dataset to separate demographic shift from compound synthesis interactions. Dynamic Threshold Calibration: The MCC analysis study highlights the need for adaptive, zero-shot threshold adaptation mechanisms to ensure reliable deepfake detectors in real world use cases.

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